

언어 모델은 가중치에 사실을 지속적으로 학습할 수 있는가?

Can a Language Model Learn Facts Continually in Its Weights?

Charles O'Neill

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초록 (Abstract)

지속 학습은 학습 이후에도 언어 모델이 새로운 사실을 계속 가중치에 기록하고 축적할 수 있다고 가정한다. 이 논문은 그 가정이 실제로 성립하는지를, 만들어진 사실을 Qwen3 계열 모델에 기록한 직후부터 이후 20 – 100 회의 추가 기록 뒤까지 추적하여 시험한다. 평가는 recall, paraphrase, application, composition, counterfactual 의 다섯 질문 유형을 사용하며, 같은 사실을 원래 모델의 prompt 에 넣은 조건을 reference 로 삼는다.

훈련 데이터의 폭 (**breadth**) 이 기록되는 지식의 종류를 결정했다. 사실 문장 하나만 반복한 bare-statement training 은 암송에 가까운 결과를 만들었다. 다양한 재진술을 사용한 study training 은 정답 결론을 직접 보여주지 않고도 recitation-to-use gap 을 27.4 점에서 5.4 점으로 줄였다. 20 회의 순차 기록 뒤 bare-statement 사실의 정확도는 1%였지만, 폭넓은 study data 로 기록한 사실은 46%를 유지했다.

행동적으로 잊힌 사실이 가중치에서 지워진 것은 아니었다. 잊힌 사실은 최초 기록이 더한 log-probability 상승의 상당 부분을 유지했고, bare-statement 조건의 오답 중 70%는 가장 최근에 기록한 사실을 포함했다. 잊힌 study fact 를 prompt 에 다시 넣으면 해당 질문 정확도가 77 – 80%로 회복되었다. 일반 능력 손상은 원래 모델과의 KL divergence 에 따라 정렬되었다. 저자들이 시험한 어느 intervention 도 이전 사실을 안정적으로 도달 가능하게 만들지 못했으며, 사실의 합성이나 장기 생존이 필요한 조건에서 context 가 weights 보다 신뢰 가능한 채널이었다.

1. 서론 (Introduction)

새 사실은 context 또는 weights 라는 두 경로로 모델에 들어갈 수 있다. context 는 즉시 사용할 수 있지만 prompt 의 수명과 context budget 에 제한된다. weight write 는 prompt 밖에 남지만, 그 지식이 다양한 질문에 사용 가능한지, 뒤의 write 를 견디는지, 다른 능력을 손상시키지 않는지가 불명확하다.

논문은 “사실 하나를 잘 외웠는가” 만 보지 않고 세 요구조건을 분리한다. 첫째, 각 write 가 단순 문장 재현이 아니라 사용할 수 있는 지식을 만들어야 한다. 둘째, 반복 write 중 일반 능력이 보존되어야 한다. 셋째, 먼저 기록한 사실이 나중에도 질문으로부터 **reachable** 해야 한다. 실험은 이 세 조건을 동일한 invented-fact 환경에서 연결한다.

2. 모델이 사실을 사용할 수 있는지 측정하기

2.1 사실 생성과 다섯 질문 유형

평가 사실은 사전 학습에서 보았을 가능성을 피하기 위해 발명한다. 예를 들어 “Zorvathine 은 영화 10 도 아래에서 녹는 금속” 같은 비정상 속성을 만든다. 한 사실에서 다음 다섯 유형의 held-out 질문을 생성한다.

1. **Recall**: 원 사실을 직접 말하게 한다.
2. **Paraphrase**: 다른 표현으로 같은 속성을 묻는다.
3. **Application**: 그 속성을 이용해 행동이나 결과를 도출한다.
4. **Composition**: 사실을 다른 지식과 결합한다.
5. **Counterfactual**: 통상 지식과 충돌하는 조건에서 새 사실을 우선해 추론한다.

질문 생성 과정 전체에서 held-out 된 문항만 downstream 평가에 쓴다. 같은 judge 와 같은 조건에서 비교 가능한 run 만 결합한다. 원래 모델에 사실을 prompt 로 제공한 성능은 weight write 가 도달할 수 있는 in-context reference 를 제공한다.

2.2 이중 채점과 entailment gap

어떤 답은 요청된 결론을 직접 말하지 않고 그 결론을 함의하는 사실만반복한다. 논문은 이를 구분하기 위해 strict와 lenient 두 채점 정책을사용한다. strict는 요청된 결론·값·선택을 명시해야 정답으로 본다. lenient는 목표를 엄밀히 함의하는 사실 진술도인정한다.

$$\text{entailment gap} = \text{lenient accuracy} - \text{strict accuracy}.$$

큰 gap은 모델이 사실을 저장했다라도 그것을 질문에 맞게 사용하는대신 암송에 머문다는 뜻이다. 두 정책은 독립적으로 판정되므로 작은 음수차이는 judge noise 범위로 취급하고 큰 차이만 해석한다.

3. 훈련 데이터의 폭이 write의 지식 종류를정한다

bare-statement training은 새 사실 문장을 직접 학습한다. study training은같은 사실을 다양한 문맥·재진술·관계로표현하지만 평가 결론을 답으로 주지는 않는다. 동일 optimization budget에서 비교하면 폭넓은 데이터가 application, composition, counterfactual을 크게 높이고 entailment gap을 줄인다.

96-step 조건에서 broader data는 application을 21점, composition을 18점, counterfactual use를 29점 높였지만, bare statement에 optimization step만더하는 것은 이 능력을 개선하지 못했다. study training은 composition 60%, bare statement는 40%, fact-in-prompt ceiling은 83%였다. 가장좋은 weight-based 방법인 context distillation은 70%였지만 in-context ceiling에는 미치지 못했다.

논문은 text objective 외에도 offline context distillation과 online context distillation, 두 KL 방향, LoRA와 full fine-tuning, RL 변형을비교한다. 목표가 달라도 training-data breadth가 write의 사용 가능성을좌우한다는 패턴은 유지된다. RL 변형은 때때로 reward를 올리면서도 사실 자체를 안정적으로 설치하지 못해 첫 요구조건에서 실패했다.

4. 지식의 종류가 순차 write 뒤 생존을좌우한다

각 사실을 adapter에 기록하고 merge한 뒤 다음 사실을 기록하는 과정을 20회, 일부 실험에서는 100회까지 반복한다. 20 write 뒤 bare-statement 사실은 1%만 유지되었고 study fact는 46%를 유지했다. study fact의 장기 retention은 100 write 부근에서 25–28% 수준으로 plateau를 보였다. factor 조건 전체에서 초기 entailment gap은 생존과 음의 상관($\rho = -0.526$)을 보였다.

이는 “처음 잘 외운 사실이면 나중에도 남는다”라는 단순 설명과 다르다. 처음 만들어진 지식이 질문에 따라 사용할 수 있는 형태인지가 후속 간섭에대한 생존을 예측한다. 하지만 broader data도 reachability 문제를해결하지는 못한다. retention을 높일 뿐, 계속되는 write에서 모든 이전사실을 안정적으로 보존하지 못한다.

5. 망각이 파괴하는 것: 저장이 아니라접근

행동 정확도가 0에 가까워진 사실의 checkpoint를 분석하면, 최초 write가더한 drift-corrected log-probability lift의 57–67%가 남아 있다. 즉 해당 token sequence의 확률 흔적이 완전히 삭제된 것은 아니다. 질문을 넣었을 때올바른 흔적에 도달하지 못한다.

bare-statement write에서 잊힌 사실에 관한 오답의 70%는 가장 최근에기록된 사실을 끌어온다. 새 write가 오래된 정보를 물리적으로 덮어쓰기만한 것이 아니라, 질문이 최신 write의 내용으로 잘

못 routing 되는 현상이다. 잊힌 study fact 의 문장을 prompt 에 다시 제공하면 77 – 80%로 회복된다. 같은 weights 안에 남아 있던 지식을 context cue 가 다시 reachable 하게 만든다.

두 사실을 각각 성공적으로 기록해도 둘을 함께 조합하는 질문에는 약하다. 이 결과도 저장량보다 on-demand retrieval 과 binding 이 병목이라는 해석과 일치한다. 논문은 이를 **question-keyed storage** 라고 표현한다.

6. 반복 write 가 치르는비용

새 사실을 기록할수록 unrelated capability 가 얼마나 손상되는지 측정한다. 손상의 크기는 원래 모델의 출력 분포에서 얼마나 멀어졌는지를 나타내는 KL divergence 와 함께 움직인다. frozen reference 를 이용한 distillation 이나 local drift penalty 는 일반 능력 보존에는 도움을 줄 수 있다.

그러나 capability 보존과 old-fact reachability 는 같은 문제가 아니다. 원래 모델과 가까운 분포를 유지해도 이전 사실을 묻는 질문이 정확한 사실로 routing 된다는 보장은 없다. 저자들이 각 write 의 gradient 와 local curvature 등 정확한 지역 측정을 이용한 intervention 을 시험했지만, incoming write 가 일으키는 간섭을 안정적으로 제어하지 못했다.

7. 생성과 간섭에 대한 인과시험

논문은 data breadth, objective, LoRA rank, model scale, write order 를 바꾸어 초기 지식 형성과 후속 간섭을 분리한다. bare-statement gap 은 4B 와 8B 모델, LoRA 와 full fine-tuning 에서 재현되었다. study data 의 어떤 개별성분이 효과를 만드는지는 완전히 분해되지 않았고, study training 이 recall 과 paraphrase 에서는 때로 낮은 이유도 남은 질문이다.

후속 write 를 bare statement 로 할지 study data 로 할지 바꾸어도 먼저 기록된 사실에 대한 간섭이 사라지지 않는다. 즉 이전 write 를 더 잘 만드는 것과 다음 write 가 가하는 충돌을 막는 것은 별도의 intervention 을 요구한다. 본문과 부록은 validity screen, judge audit, step 수, rank, model size, cue phrasing 에 대한 절제 실험을 제공한다.

8. 관련 연구 (Related Work)

논문은 knowledge editing, continual fine-tuning, catastrophic forgetting, model merging, distillation, parameter-efficient adaptation, in-context learning 을 비교한다. 기존 연구가 한 번의 edit 성공률이나 general benchmark 보존을 주로 측정했다면, 이 연구는 동일 사실을 처음 write 한 순간부터 다수의 후속 write 뒤까지 추적하고 context channel 을 고정 ceiling 으로 둔다.

9. 논의 (Discussion)

실험은 발명된 소수 사실, Qwen3 계열, 특정 adapter merge schedule 과 자동 judge 에 한정된다. 모든 형태의 장기 pretraining 이나 대규모 continual corpus 로 곧바로 일반화할 수 없다. log-probability 흔적이 남았다는 결과는 완전한 의미 표현이 그대로 저장되었다는 증거가 아니며, 관측한 checkpoint 와 질문 집합 안에서 access failure 가 유력하다는 증거다.

결론은 weight update 가 언제나 무용하다는 것이 아니다. broader training data 는 훨씬 사용 가능한 지식을 만들고, frozen reference 나 KL 제약은 capability drift 를 줄인다. 그러나 여러 사실의 composition 과 뒤이은 write 생존을 동시에 요구할 때, 이 논문이 시험한 weight-write 방법 중

어느 것도 context reference 의 신뢰성에 도달하지 못했다. 따라서 저자들은 신뢰성이 필요한 사실 채널로 context 를 유지하고, weight write 는 그 운영 경계를 명확히 측정해야 한다고 결론짓는다.

부록 안내

원문 부록에는 dataset certification, judge 의 strict/lenient audit, training step·rank·model scale factorization, 100-write 장기 run, cued phrasing, reconstruction 과 log-probability 분석, local-control intervention 의 상세 결과가 있다. 뒤의 원문 보존 부록은 모든 figure·table·equation 과 수치표를 v2 PDF 그대로 실는다.

번역·검증 고지

이 문서는 분석이나 평가를 추가하지 않는 것을 원칙으로 하며, 원문의 주장 강도와 수치를 유지하도록 작성되었다. 한국어 초안 작성과 구조 검수에는 Codex GPT-5.6 을 사용했고, 고정 source hash·원문 보존 부록·역대조 표를 통해 검증했다. 불일치가 있을 때에는 원문이 우선한다.

원문 구조·그림 보존 부록

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Can a Language Model Learn Facts Continually in Its Weights?

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Abstract

Continual learning promises a language model that keeps acquiring knowledge after training, with each new fact written into its weights. Whether weight writes can support accumulation remains undecided. We follow invented facts written into Qwen3 models from creation through sequences of twenty to one hundred later writes, using held-out questions of five types, with the original model given the fact in its prompt as the reference. Across these experiments, the breadth of the training data determines the kind of knowledge created. Bare-statement training produces recitation, while diverse restatements reduce the recitation-to-use gap from 27.4 to 5.4 points without showing the model a conclusion. This difference carries into later writes: after twenty sequential writes, bare-statement facts retain 1% accuracy while facts written from broad study data retain 46%. We also find that facts can be behaviourally forgotten without being erased. Forgotten facts keep most of the log-probability added by their write, and under bare-statement training 70% of wrong answers about them contain the most recently written fact. The same writes barely degrade the model's use of facts in context, and a forgotten study fact supplied in the prompt recovers to 77–80% on its questions. These results describe knowledge that is stored but question-keyed: later writes redirect the questions that reached it. Damage to unrelated abilities tracks KL divergence from the original model, and the later writes cause interference regardless of how the earlier fact was stored. Broad data can create usable knowledge, and a frozen reference can preserve capability, but no intervention we tested, including those built on accurate local measurements of each write, keeps earlier facts reachable. When facts must be composed or survive later writes, the reliable channel is context rather than the weights.

1. Introduction

A language model can hold a new fact in its context or its weights. Context makes the fact immediately usable, but only for the life of the prompt. Continual learning asks the weights to acquire facts after training and retain them through further writes. To understand whether they can, we characterise the object that a write creates: what kind of knowledge it contains, whether later writes preserve it, and what remains after questions about it fail. This turns catastrophic forgetting into a property of the written object, measured against the same content placed in context.

Fine-tuning learns unknown facts slowly, and learned facts fail reversals and multi-hop use that the same facts support in a prompt (Gekhman et al., 2024; Berglund et al., 2024; Lampinen et al., 2025). Diverse paraphrases make facts more extractable (Allen-Zhu and Li, 2024), while distributional drift predicts forgetting under further training (Shenfeld et al., 2025). These observations lack a common account of why some writes produce usable knowledge, why some survive, and what a forgotten fact leaves behind.

We build that account with invented facts, held-out questions of five types, and two fixed references: the original model and the same model with the fact in its prompt (Figure 1). Following each fact from creation through twenty to one hundred later writes lets us connect what a write creates to what later survives. Training-data breadth determines whether the model learns recitation or stated conclusions, and this difference predicts retention. When a fact eventually fails every question, checkpoint reconstructions still find most of its write's log-probability lift in the weights, and under bare-statement writes the questions that once reached it return the newest write's content instead. The same access problem is present before any overwriting: two individually usable written facts largely cannot be used together because the model cannot retrieve them on demand. By contrast, context remains usable through the same writes, and questions about a forgotten fact succeed again once its statement is supplied.

Code, training conditions, and run records: <https://github.com/basetenlabs/cortex>. Evaluation datasets: <https://huggingface.co/datasets/baseten/cortex>.

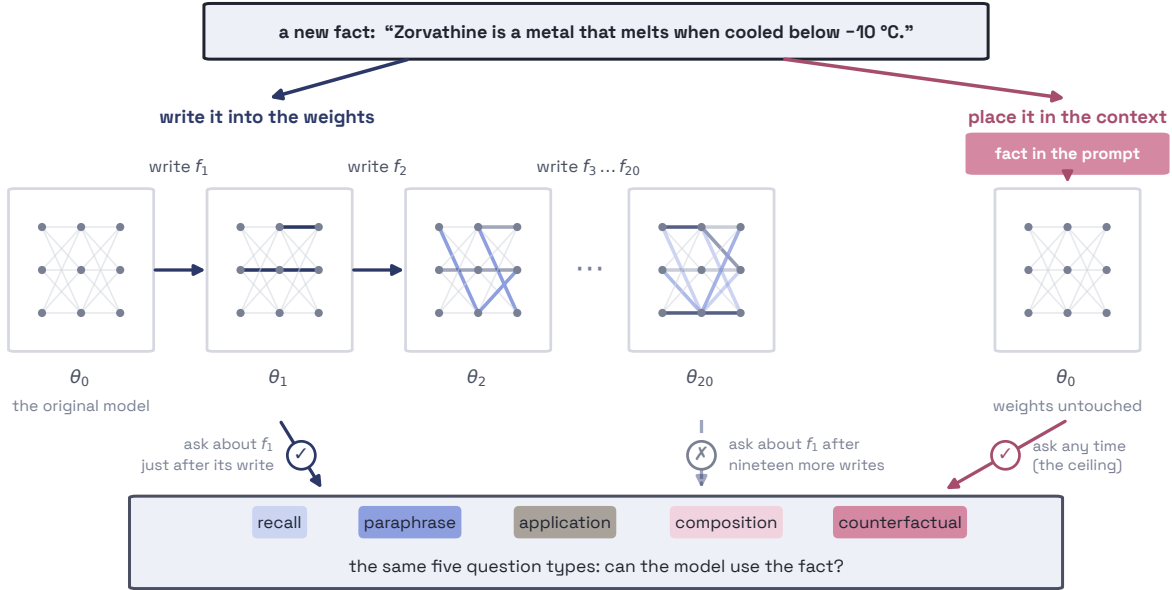


Figure 1: The setup. A new fact reaches the model through two channels. Written into the weights, each fact trains an adapter that is merged before the next fact arrives; placed in the context, the fact accompanies every question and the weights stay untouched. Both channels face the same five question types (§2), asked immediately after a fact’s write and again after later writes. The marks preview the finding: a written fact answers just after its write and often not after twenty more, while the context channel keeps answering.

Together, these results separate three requirements for continual writing. Each write must create usable knowledge, general abilities must survive accumulation, and earlier facts must remain reachable. The first two appear tractable: broad data creates knowledge the model can apply, while a frozen original model or a penalty on local drift can protect general abilities. Reachability remains unsolved across supervised fine-tuning and both offline and online distillation; a reinforcement-learning variant fails at the first requirement, occasionally gaining reward without installing the fact. Yet on every schedule where the comparison passes its validity screen, in-context use erodes no faster than general ability, leaving context as the reliable channel when facts must be composed or outlive later writes.

Our five contributions are:

1. Broad training data makes written knowledge usable. Diverse restatements reduce the recitation-to-use gap from 27.4 points to 5.4 without showing conclusions, and the bare-statement gap replicates at 4B and 8B scale and under LoRA and full fine-tuning (§3, §7).
2. Keeping earlier facts remains unsolved, and the kind of knowledge predicts survival. After twenty sequential writes, bare-statement facts retain 1% and study facts retain 46%; the entailment gap predicts survival across the factorial ($\rho = -0.526$), and study retention plateaus at 25–28% by one hundred writes (§4).
3. Forgetting destroys access rather than storage. Forgotten facts keep 57–67% of their write’s drift-corrected log-probability lift, under bare-statement writes their failures carry the newest write’s content, and questions about them succeed again when their statements are supplied in context (§5).
4. General abilities can be preserved because capability damage orders by KL divergence from the original model ($\rho = 0.83$ over twelve conditions and 0.946 in the factorial). A frozen teacher writes facts at near-zero cost, while distillation through accumulated writes is the most damaging condition (§6).
5. The incoming write causes interference, which resists local control. Study rather than bare-statement later writes improve retention by 37.6 points, while the stored-write method has no detectable effect; a linearised

predictor captures the immediate effect of an update ($\rho = 0.795$) but not the full trajectory ($\rho = -0.258$) (§7).

§2 describes the evaluation. Model-generated data trains its conditions, model-written questions test them, and model judges score the open answers, so the instrument’s validity must be demonstrated. Its certification requirements and dual grading policy support every later result.

2. Measuring whether a model can use a fact

Every claim in this paper depends on deciding whether a model can use a fact it did not previously know, rather than merely repeat it. We test five forms of use against a floor from the original model and a fact-in-prompt ceiling. Appendix A gives the complete training details and a glossary of conditions.

All experiments use Qwen3-4B with a frozen base and LoRA adapters unless stated otherwise (Hu et al., 2022). The primary evaluation contains 247 one-sentence facts about invented entities; the original model demonstrably fails their questions (the floor screen below). Most facts contradict a familiar default. For example, “*Zorvathine is a metal that melts when cooled below -10°C .*” A correct answer must therefore come from the new fact rather than general knowledge. Appendix D varies the strength of this conflict.

Every fact receives questions of these five types (Table 1), with gold answers and accepted aliases. Recall and paraphrase test recitation; application requires one-step use; composition combines the fact with unstated world knowledge; and counterfactual questions force a choice between the written fact and the model’s default. The two anchors run on every fact: the original model, with no fact in view, forms the floor, at 1–7% across question types, and the same model with the fact in its prompt forms the ceiling, at 75–99% under the strict policy below.

Scoring combines deterministic checks with two model families. The checker handles exact and alias matches and is tested against negation and numeric sign flips: “it will *not* turn blue” cannot match *blue*, and “below -10°C ” cannot match “below 10°C ”. Remaining answers go to an item-level model judge, and passes must survive a second judge from another family. Errors score as incorrect, and every compared condition uses the same pipeline. Uncertainty is a bootstrap 95% confidence interval over facts; comparisons use paired per-fact or per-triple differences, and multi-seed conditions report each seed and their mean. Confirmatory contrasts and their thresholds were fixed before the outcome runs, and post-hoc analyses are labelled as such. Exclusions made during screening and judging-validity checks leave 236–247 facts per contrast; each analysis states its n .

2.1 Validity requirements and certification

Generated facts, training data, questions, and judgments create six recurrent failure modes, listed with their bias directions in Table 2. Three determine the design. A shared generator can teach the evaluation’s phrasing through the training data. An asymmetric harness constraint can punish one answer style through a length cap, numeric normalisation, or a checker that misses negation. A judge that abstains on a non-random subset can also reshape a comparison even when abstentions fail closed.

We certify the instrument before scientific use. The gate requires a floor below 20%, a ceiling above 80%, truncation below 5% per condition, an adversarial audit of answer leakage, and audits of judged passes as well as failures. The version used here passes at floor 3%, ceiling 84%, leak 1/30, and pass-audit 29/30. Two conditions exceed the truncation bar and are disclosed rather than gating: the original-model floor (23%) and bare-statement

Table 1: Five question types for the Zorvathine fact. Questions are written by a different model family from the facts and training data, reveal no answer, and are retained only when the original model fails.

type	example question	expected answer
recall	State Zorvathine’s unusual temperature property.	It melts below -10°C .
paraphrase	What phase change occurs when Zorvathine becomes colder than -10°C ?	It melts.
application	How would you liquefy a solid sample of Zorvathine?	Cool it below -10°C .
composition	Could a solid Zorvathine container hold liquid nitrogen without melting?	No; the cold would melt it.
counterfactual	Would cooling solid Zorvathine make it melt or freeze?	It would melt.

training at 24 steps (11%). Truncation forces failures, so both deflate the floor side of comparisons (Appendix A). Downstream, *held out* means held out from the entire generative process, and comparisons only join conditions scored by the same judge.

2.2 The dual grading policy and entailment gap

Answers that state an underlying fact without drawing the requested conclusion admit two defensible grades. We therefore score every answer twice. The *strict* policy requires the requested conclusion, value, choice, or name; the *lenient* policy also credits a fact that strictly entails the target. Use-type claims anchor on the strict policy, while the policies nearly coincide on recall and paraphrase. The primary judge is GPT-5.4-mini, configured with reasoning off at temperature zero, and it returns a decision on every item. Because the judging pipeline is part of the instrument, the certification was repeated under this judge and the dual policy, and on the final records the instrument again clears its gate: floor 4%, in-prompt ceiling 91% lenient, and strict pass-audit 30/30.

The difference between the policies measures how often a method stops at recitation instead of stating the conclusion:

$$\text{entailment gap} = \text{lenient accuracy} - \text{strict accuracy}. \quad (2.1)$$

This is the recitation-to-use gap of the abstract and introduction. The two policies are judged independently rather than nested, so small negative gaps occur, and differences of a few points sit within judge noise; we interpret only large gaps. Section 3 shows that the gap differs sharply across training methods. We treat it as a property of the written object, not a grading artefact to tune away.

3. Training-data breadth sets the kind of knowledge a write creates

The first requirement of continual writing is that each write create usable knowledge. To determine which methods meet it, we compare training methods at matched optimisation budgets and use the entailment gap to measure the kind of knowledge they create (§2.2). Across objectives, ranks, and model scales, bare-statement training is the only method that leaves a large gap. Broader training data closes it, and the factorisation of §7 identifies that breadth as the causal variable.

3.1 Training-data breadth

We begin with two conditions on the 247-fact evaluation. *Bare-statement* training uses the fact sentence in two trivial framings; *study* training uses 24 generated paraphrases, question–answer pairs, worked implications, and contrasts with the default the fact violates. Bare-statement budgets are 24, 96, and 192 steps, and study budgets are 24 and 96. The heaviest bare-statement budget, 192 steps on a single sentence, is deliberately severe; the lighter-budget control of §4.1 and the matched-budget factorisation of §7 show that the conclusions do not depend on that operating point.

The five-type average at 96 steps depends on the grading policy. Under the strict policy used for use-type claims, study training wins by 10.1 points (95% CI [+7.0, +13.3], paired over 243 facts), while bare-statement training wins under the lenient policy by 13.6 points. The question types explain the reversal (Figure 2).

The two conditions trade off recitation and use. Bare-statement training reaches 97% recall and 95% paraphrase within 24 steps and saturates by 96, whereas study training reaches 88% recall and 83% paraphrase at 96 steps, trailing bare-statement training at its 96-step saturation by paired differences of 11 and 13 points. The pattern reverses when the written fact contradicts the model’s default. Bare-statement accuracy stays at 21–23% from 24 to 192 steps, while study training reaches 45–50%, a paired gain of 29 points at 96 steps. Further optimisation on the statement therefore improves recitation without substituting for varied data.

The reversal between grading policies makes this distinction visible (Figure 3). The entailment gap is 22–42 points on bare-statement use questions, compared with 1–5 points after study training and 1–6 points with the fact in the prompt. Lenient grading credits a recited premise, while strict grading requires the requested conclusion, so the large bare-statement gap measures knowledge that stops one step short of use.

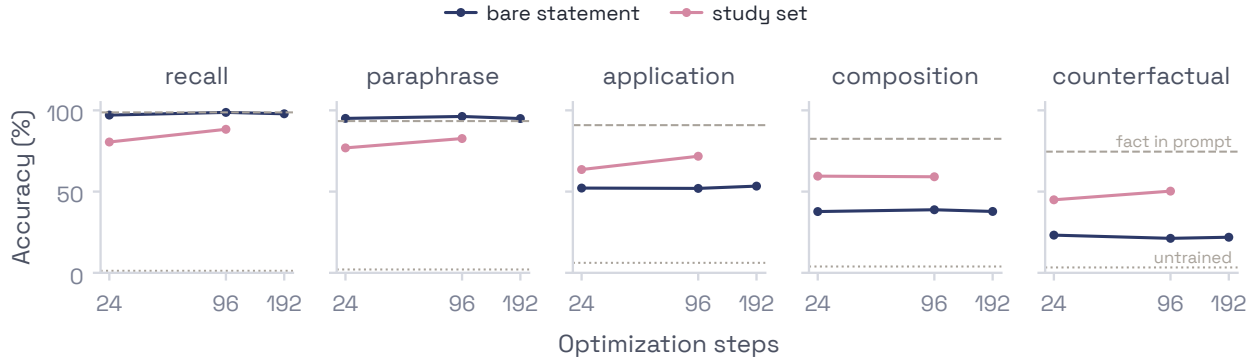


Figure 2: Strict accuracy by question type as the optimisation budget grows for bare-statement training (indigo) and study training (rose). The dashed line is the fact-in-prompt ceiling and the dotted line the floor from the original model. Counterfactual accuracy stays at 21–23% under bare-statement training and reaches 45–50% under study training; composition is 39–41% against 59–60%, below the 83% ceiling.

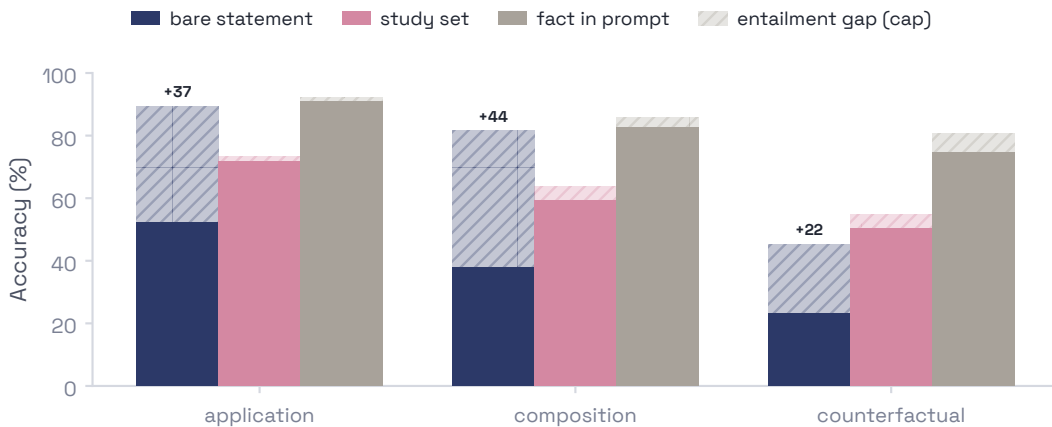


Figure 3: The entailment gap on the primary evaluation. Solid bars show strict accuracy and light caps extend them to lenient accuracy. The cap is the share of answers that state an entailing fact without giving the requested conclusion.

Broader data also improves composition, although no method reaches the ceiling. Study training reaches 60% against 40% for bare-statement training and 83% with the fact in the prompt, a paired gain of 18 points. The best weight-based method, context distillation (a distribution-matching objective defined in §3.2), reaches 70%, placing it 10.8 points above study training and 13.1 points below the ceiling. Fine-tuning can therefore improve composition, but no weight-based method we tested matches in-context composition.

Taken together, the comparison separates what optimisation and data provide. At 96 steps, broader data improves application by +21 points, composition by +18, and counterfactual use by +29, while extra bare-statement optimisation improves none of them. We have not isolated which ingredient of the study data matters, and why study training scores worse on recall and paraphrase questions remains unexplained. These comparisons use uncued questions, which never restate the training phrasings; cued phrasings can supply the missing step themselves and collapse use back to recitation, a boundary measured in Appendix D.4.

3.2 Training objectives

The first comparison changes the training data under a single text objective, but a write can instead target a distribution. We therefore compare bare-statement and study training with offline context distillation and online context distillation in both KL directions. Each method writes the same 40 certified facts into a fresh rank-16

adapter for 192 steps, with three seeds. The steps are matched, but generation and token throughput differ, so this remains an operating-point comparison.

Writing a fact s with supervised fine-tuning maximises the likelihood of target text y given prompt x over the fact’s training set $\mathcal{D}_s$:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,y) \sim \mathcal{D}_s} \sum_t \log \pi_\theta(y_t | x, y_{<t}). \quad (3.1)$$

For context distillation, the teacher $\pi_T(\cdot | x, y_{<t}) = \pi_0(\cdot | s \oplus x, y_{<t})$ is the original model with the fact in its prompt. The student does not see the fact and matches the teacher token by token. Offline distillation matches the student to this distribution on sequences sampled once from the teacher,

$$\mathcal{L}_{\text{offline}}(\theta) = \mathbb{E}_{x, y \sim \pi_T} \sum_t \text{KL}(\pi_T(\cdot | x, y_{<t}) \| \pi_\theta(\cdot | x, y_{<t})), \quad (3.2)$$

while online distillation samples $y \sim \pi_\theta$ and uses the same token-level divergence in the forward direction ($\text{KL}(\pi_T \| \pi_\theta)$) or reverse direction ($\text{KL}(\pi_\theta \| \pi_T)$). The reverse direction is the on-policy-distillation objective of [Lu and Thinking Machines Lab \(2025\)](#).

Across the five methods, bare-statement training alone has a large entailment gap, at 26 points, while the other four lie at 2–4 points. The contrast between the bare-statement gap and their mean is +22.8 points (95% CI [+17.4, +28.2]). Richer supervision therefore closes the gap whether it comes from broader text data or a distributional target.

Once the gap is within judge noise, changing the KL direction adds no further effect. Our prediction that mode-seeking reverse KL would create more committed knowledge fails: the forward and reverse online-distillation gaps are 4.3 and 3.0 points. On-policy sampling does, however, improve strict accuracy at this operating point. Online distillation reaches 77–78%, compared with 71–72% for study training and offline distillation and 61% for bare-statement training; every method remains above 50%. Student-sampled sequences buy seven points over fixed teacher samples, and §4 tests whether that advantage survives later writes.

3.3 A factorial over objective, data, and update method

Because the method comparison changes the source of the targets and the loss together, we next cross four objectives (SFT, offline distillation, and online distillation in each KL direction), bare-statement or study data, and LoRA or full fine-tuning. The resulting sixteen conditions each have three seeds and forty facts, and fifteen yield numeric results. Reverse-KL distillation on bare statements with LoRA is censored because 9.4% of its outputs loop until the generation cap, consistently across seeds; raising the cap extends the loop.

The data effect survives this separation: study data reduces the conclusion-type entailment gap in every reportable pair (Figure 4a). The reductions are 36.4 points (95% CI [31.7, 41.0]) for full-parameter SFT, 31.2 for LoRA SFT, 18.9 for full offline distillation, 14.0 for full forward-KL online distillation, and 3.6–9.8 for the remaining pairs. Four of seven pairs exceed ten points, and five exclude zero. Because four pairs use learning rates tuned by condition, the factorial gives an association across operating points rather than a one-variable causal effect. The controlled factorisation in §7 supplies that identification; here the result shows that the objective label alone is too coarse.

A GRPO condition with a programmatic answer-match reward received almost no signal because the base model never produced the target counterfactuals, and the few reward gains did not install the fact; we leave reinforcement-learning objectives that can write new knowledge to future work.

3.4 Scale and update method

The same entailment gap appears at a larger model scale (Figure 5a). At Qwen3-8B, the bare-statement gap remains 26 points and the headline contrast is +22.9 points, compared with +22.8 at 4B.

Full fine-tuning requires matched writing strength. At learning rate 10^{-5} it produces a small bare-statement gap because it barely writes the fact: lenient accuracy is 35%, against 87% for LoRA. We swept the learning rate on five held-out facts and selected 3×10^{-5} because it matched LoRA’s writing strength rather than any target

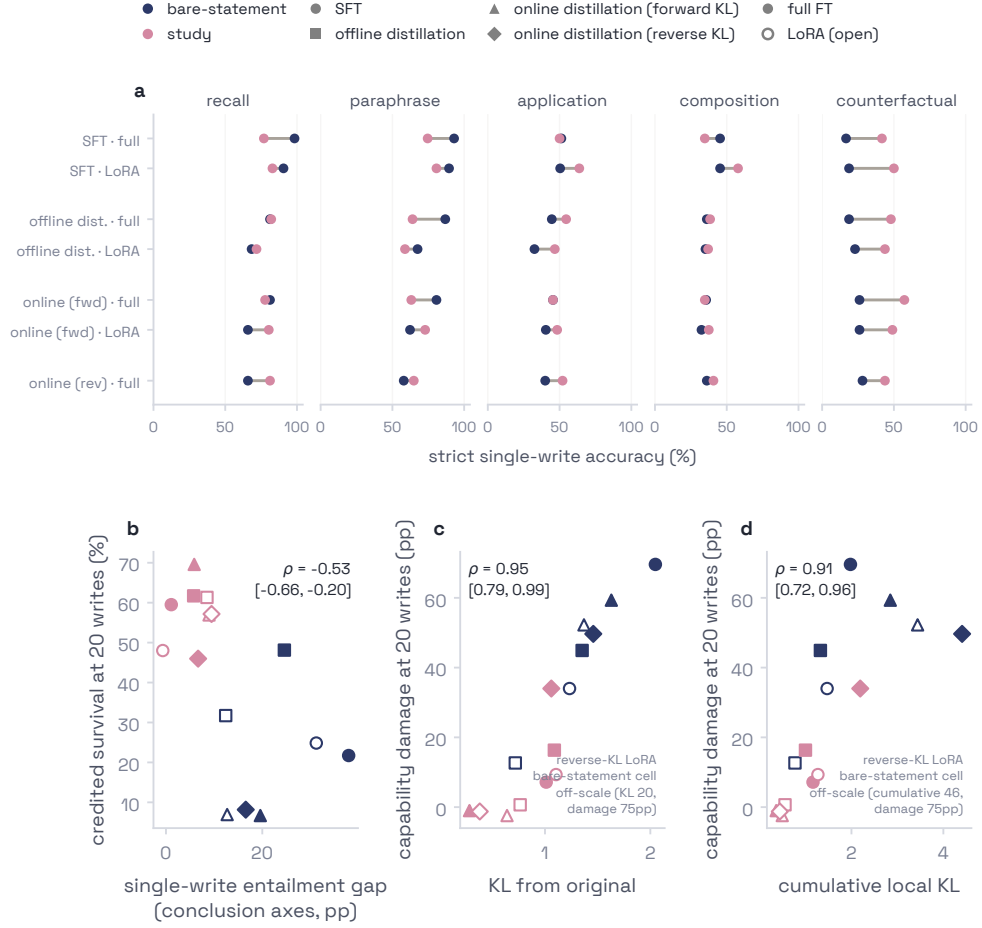


Figure 4: The factorial over objective, data, and update method. **a**, strict accuracy after one write, with each connected pair showing the bare-statement-to-study change. **b**, the conclusion-type entailment gap against survival after twenty later writes. **c–d**, capability damage against endpoint KL from the original model and cumulative local KL. Colour denotes training data, marker shape the objective, and fill the update method; intervals are 95% condition-clustered bootstrap intervals. The correlations include all sixteen conditions, with the censored condition annotated off-scale.

gap; the largest rate destabilised the study condition. At the matched setting, full fine-tuning reproduces the gap: 28 points for bare-statement training and 4 for study training, a +24.0-point contrast (95% CI [+20.9, +27.2]). Narrow supervision, not the number of trainable parameters, creates the gap.

4. The kind of knowledge governs what survives continual writing

The previous section measures what each method writes. We now follow that knowledge through later writes and find that its kind predicts survival: recitation disappears, while stated conclusions survive at far higher rates. The result holds across two evaluations and out to one hundred writes.

4.1 Retention under sequential writing

We repeat the five-method comparison of §3.2 in a sequential setting. Each method writes twenty facts one at a time, merging each fact’s adapter into the model before the next fact trains, at the same 192 steps per fact, with three seeds. After the twentieth write we re-evaluate every earlier fact (Figure 5b).

Facts written by study training retain 46%, while facts written by bare-statement training retain 1%, a paired difference of 45.6 points (95% CI [38.8, 52.6], $n = 57$ fact–seed cells). The collapse is not a failure to write: bare-statement training answers 65% of questions correctly before any later writes arrive. Nor is it degenerate output. Inspecting the failures, the model gives coherent answers drawn from other written facts. The fragility is also not collateral damage from a heavy optimisation budget. At a lighter budget of 24 steps per fact, which leaves the held-out capability tests at 54% (against 3–37% at the full budget and roughly 80% before any writing), bare-statement facts still retain only 6%, so capability and retention dissociate.

The comparison also shows an on-policy advantage. Online context distillation retains 27–32% against 14% for offline distillation (+15.8 points), and this is not an artefact of writing strength, since the reverse-KL variant writes less strongly than offline distillation yet retains more than double. One caveat changes the reading: the entire distillation family partially degenerates by the twentieth write (35–46% looping output, capability down to 42–49%), so the on-policy advantage is real in direction but modest in a degraded regime. Of the five methods as run here, study training is the only one that neither degenerates nor collapses, and it still loses more than half of its facts; §6 shows that a frozen teacher repairs the distillation conditions.

The factorial supplies a cross-method test. For each of the three conclusion question types, we compare a condition’s single-write entailment gap with survival after twenty writes. Across 45 such pairs, Spearman $\rho = -0.526$ (95% CI [−0.657, −0.197]; Figure 4b). Collapsing each condition to its mean gives $\rho = -0.675$. Sensitivities on a second question source and on the pooled set agree in sign. Because the analysis plan did not anticipate censoring, this is a complete-case analysis of fifteen conditions rather than the planned sixteen. The censored condition’s repetitive output grows from 9.4% after one write to 85.2% after twenty, so it is itself a method collapse rather than a benign omission.

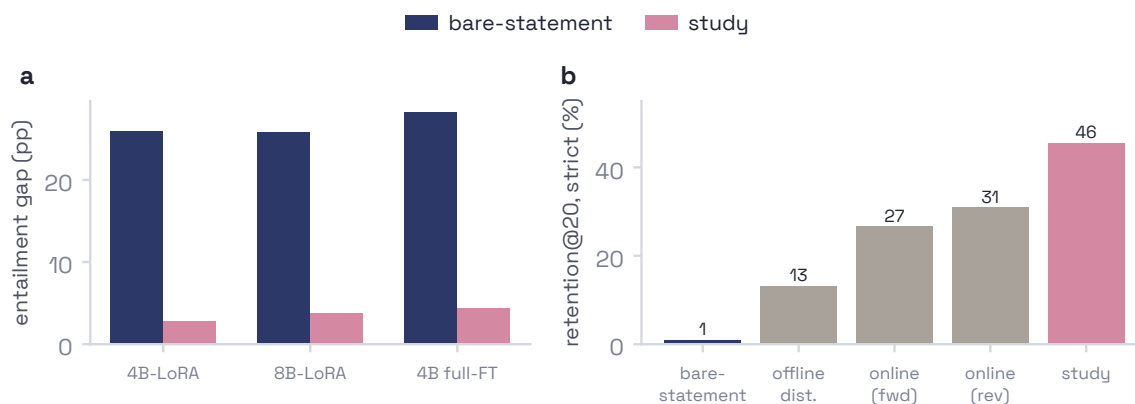


Figure 5: **a**, the entailment gap for bare-statement and study training in three configurations; the gap follows the training data, not rank or scale (§3.4). **b**, retention of earlier facts after twenty sequential writes, strict, with counterfactual questions excluded. Recitation dies; half of the stated conclusions survive.

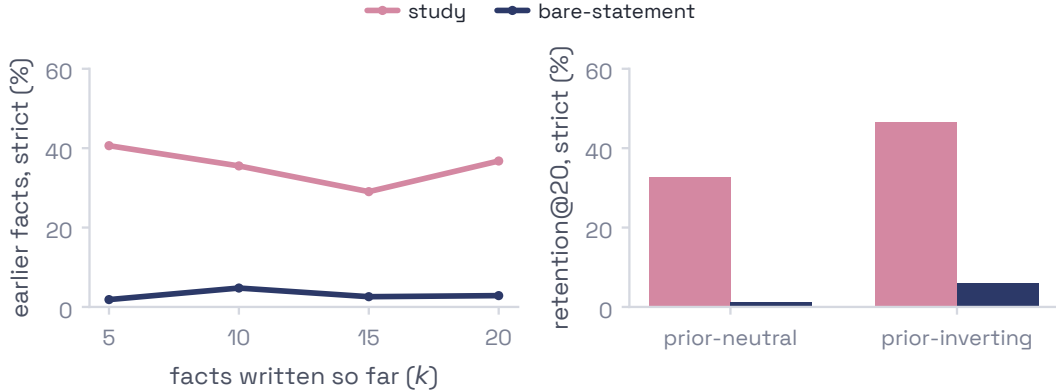


Figure 6: Retention on the prior-conflict evaluation. Left: strict accuracy on earlier facts after k writes, pooled over tiers and seeds. Facts written from bare statements are near zero by $k = 5$. Right: retention after all twenty writes by prior tier; prior-inverting facts survive better than neutral ones under both methods.

4.2 Retention on the prior-conflict evaluation

A second experiment asks whether the result depends on the fact itself (Appendix D). Four sequential conditions cross bare-statement and study training at 96 steps per fact with prior-neutral and prior-inverting facts, each run with three seeds over twenty writes. Contrasts are strict and paired. Counterfactual questions are excluded from the primary endpoints because their cue sensitivity was still under test (Appendix D.4).

The same separation appears on this evaluation. Facts written by study training retain 31–49% at $k = 20$, while facts written by bare-statement training retain 1–7%, a paired difference of 35.7 points (95% CI [29.8, 41.9], $n = 114$ fact–tier–seed cells) at an identical optimisation budget (Figure 6). Weaker writing does not explain this: bare-statement training writes these facts at 59% strict accuracy to study training’s 72%, so normalised survival still differs by a factor of about six. The collapse is immediate, with bare-statement facts at 2% by $k = 5$. We also find no decay into recitation. The lenient–strict gap of retained facts is 0–2 points at every checkpoint under both methods, so a fact either survives as a stated conclusion or disappears, with no intermediate stage as recitation-only knowledge.

The prior effect reverses our prediction that facts contradicting the model’s prior decay faster. We expected inverting facts to retain at least 10 points less than neutral ones; every re-analysis leaves the estimate at least 22 points away from that mark, and the direction reverses: neutral facts retain 12.3 points less than prior-inverting ones (95% CI [−21.3, −2.8], $n = 19$ paired triples), under both training methods. The reversal remains uncertain: a per-triple sign test gives $p = 0.064$, the study-only interval touches zero, and the two tiers use disjoint question sets that were not difficulty-matched. It is, however, stable under leave-one-out, negative on every shared question type, and it runs opposite to the writing-strength confound: inverting facts are written more weakly than neutral ones, which would predict the reverse ordering. Our working interpretation, post hoc, is that distinctiveness protects: a fact that contradicts the prior occupies unusual territory and collides less with subsequent writes than a bland fact does. Alternatives such as question-difficulty asymmetries between the tiers remain open, and the crossed experiments of §7 do not resolve them.

4.3 One hundred writes and periodic consolidation

We extended the prior-conflict experiment to one hundred sequential writes. Study retention declines to a 25–28% plateau rather than to zero (Figure 7).

The same separation holds across methods at that horizon (Figure 8). In a hundred-write extension of the factorial covering SFT and offline distillation with bare-statement or study data and LoRA or full-parameter updates, bare-statement SFT ends at 8.6–11.7% retention against study SFT’s 33.0–37.7%, and offline distillation shows the same data effect (25.0–29.4% against 38.5–43.7%; ranges span the update methods). The training-data separation therefore persists under both objectives and update methods. This extension is descriptive: its scope was reduced

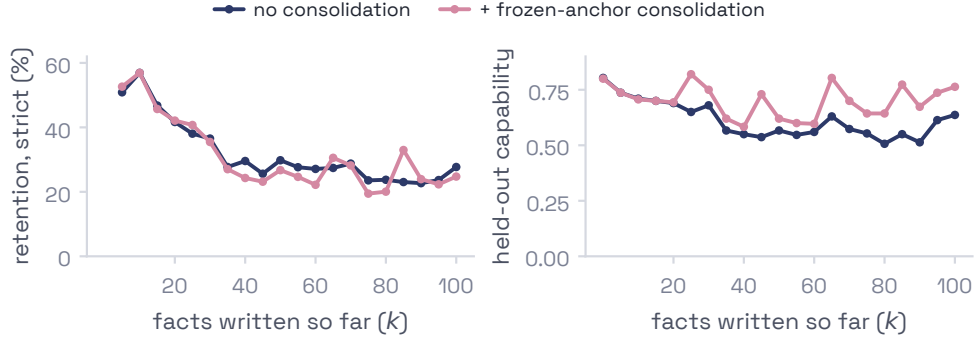


Figure 7: One hundred sequential study writes on the prior-conflict evaluation. Left: retention decays to a 25–28% plateau, and periodic consolidation into the original model does not lift it. Right: consolidation does preserve the held-out capability tests, recovering them at each pass.

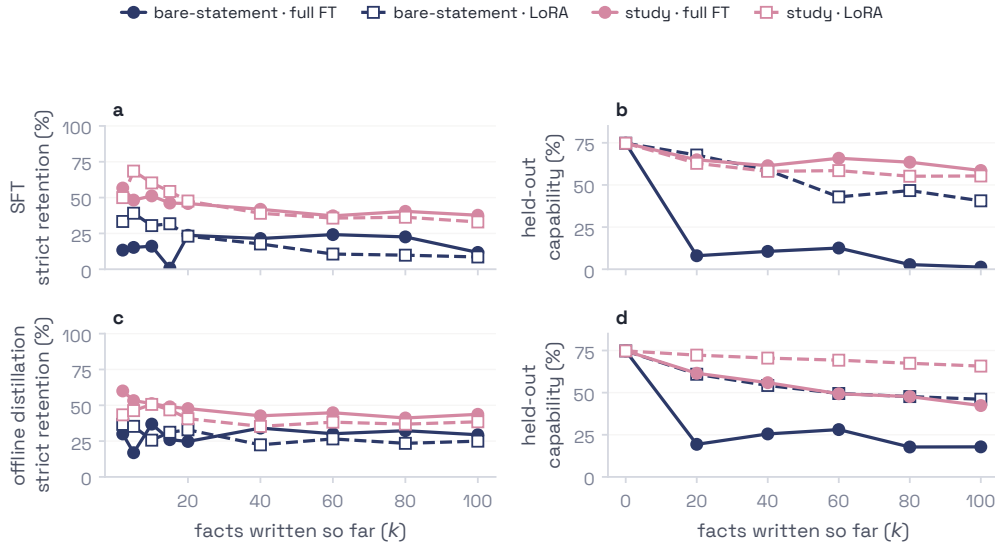


Figure 8: One hundred sequential writes across the reduced factorial (three seeds per condition; descriptive, with pooled point estimates and no uncertainty bands). Left: strict retention of earlier facts; the bare-statement/study separation persists to $k=100$ under both objectives and both update methods. Right: the held-out capability tests over the same runs.

after partial records existed, its endpoint contrasts were chosen after the fact, and its learning rates are tuned per condition, so we read trajectories and endpoint ranges rather than inferential contrasts (details in Appendix B.3).

We then test the mitigation suggested by §6: every twenty writes, consolidate all facts so far into a fresh copy of the original model by batch distillation. Consolidation does not help retention (25% against 28% at $k = 100$, within noise), and the passes are not simply losing a race with the writes between them: five writes after the first pass, retention is 41% against the unconsolidated run’s 38%, so the pass fails to restore reachability even immediately. General abilities, in contrast, recover at every pass and stay systematically above the unconsolidated run (+12 points at $k = 100$; Figure 7, right). Consolidating into the original model is a capability safeguard, not a retention fix. This is consistent with the addressing account of §5, in which a pass restores the output distribution while the routes to earlier facts stay captured; why re-distilling every fact from its statement fails to re-key them is a question our experiments do not answer.

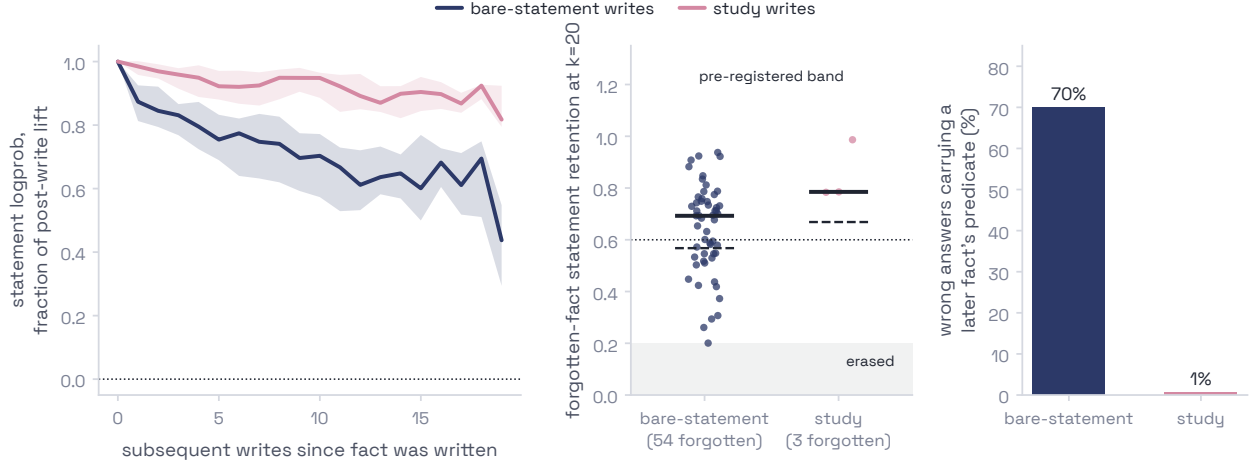


Figure 9: Storage survives behavioural forgetting. Left: the written statement’s log-probability, as a fraction of the lift its own write produced (Eq. 5.1), stays far above the pre-write floor across all subsequent writes. Lines are medians and shading the interquartile range over facts, for three seeds. Middle: facts that fail every question at $k=20$ still retain most of their statement lift. Solid lines are medians, dashed lines subtract the drift control, and the shaded region marks where genuine erasure would sit. Right: the fraction of forgotten facts’ wrong answers that contain the most recently written fact’s content.

5. What forgetting destroys: access, not storage

Failed questions do not distinguish erasure from lost access. We separate these possibilities by probing storage directly, testing joint use before overwriting, and restoring forgotten facts through context. Across all three tests, a stored trace survives while access fails.

5.1 Storage after behavioural forgetting

We repeat bare-statement and study training for twenty sequential writes and three seeds while saving every adapter. Reconstructed checkpoints agree with the original measurements within 10^{-2} nats per token. We then track the probability assigned to each written statement from before its write to the end of the sequence. For a fact with statement s written at step j and probed after k subsequent writes, we summarise storage by the retained fraction of the write’s own log-probability lift,

$$R(k) = \frac{\log p_{\theta_{j+k}}(s) - \log p_{\theta_0}(s)}{\log p_{\theta_j}(s) - \log p_{\theta_0}(s)}, \quad (5.1)$$

where θ_0 is the model before the fact was written and θ_j the model just after. $R = 1$ means the write’s lift is fully retained; $R = 0$ means erasure back to the pre-write prior. A drift control, the same quantity computed on never-written statements, corrects for the elevation that any statement receives from the surrounding writes. Facts whose write produced no measurable lift are excluded from $R(k)$, and we report medians over facts.

Facts that fail every strict question at $k=20$ still hold a median 69% (bare-statement) and 79% (study) of the log-probability lift their write produced, or 57% and 67% after the drift correction. No fact in either condition approaches the erased floor. The probe reads only the statement’s likelihood, but for study facts the write demonstrably installed more than a string, because use questions passed before the later writes arrived; the statement’s lift is the measurable trace of that object. The fact therefore remains measurably present after its behavioural loss (Figure 9).

What changes instead is where the questions land. Among wrong answers about forgotten facts, the bare-statement model attaches the most recently written fact’s content to the queried entity in 70% of failures; the study model almost never does (1%). Asked about the first fact it wrote, the model answers with the twentieth.

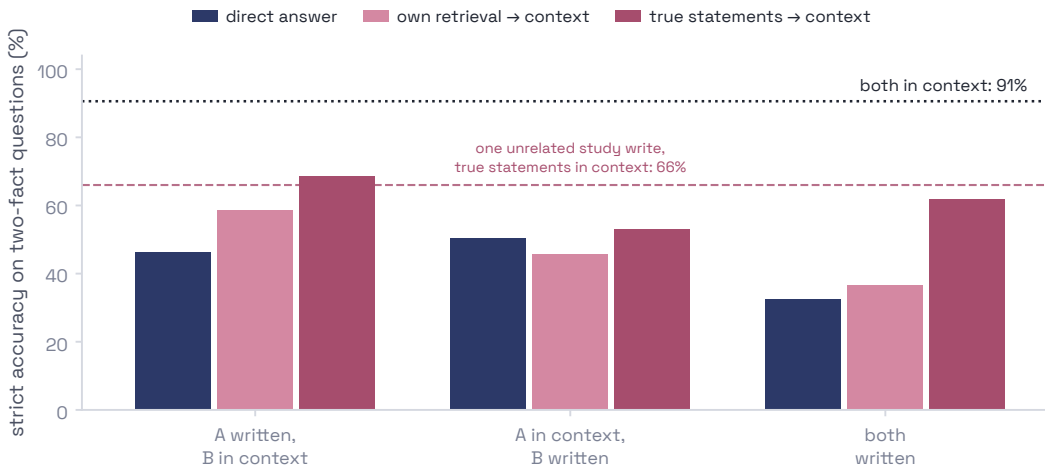


Figure 10: Two written facts largely cannot be used together. Bars show strict accuracy on questions requiring both facts of a pair, by placement. Within each placement, the model answers directly, answers after its own retrievals are placed in context, and answers with the true statements placed in context. The dotted line is the both-in-prompt reference; the dashed line is the same true-statements-in-context measurement on a model that wrote one unrelated fact.

Reaching a forgotten fact is also harder than learning a fresh one. Re-training one forgotten study fact to criterion consumed the full step budget, whereas a never-written control fact took eight steps. Slow relearning is the opposite of the classical savings effect, in which stored but inaccessible memories relearn faster, and it is what address capture predicts: retraining must displace the newer content that now owns the fact’s questions, not restrengthen a faded trace. The comparison is suggestive rather than conclusive. Only three study facts were forgotten in total (their scarcity is the survival result of §4.1), and the same probe is uninformative in the bare-statement condition, whose model after twenty writes is too damaged to learn any fact to criterion. A second probe, re-testing forgotten facts with cued questions, gave study 3/3 and bare-statement 0/54; we exclude it from quantitative use because its questions share their contrast form with the study training data.

5.2 Joint use and self-retrieval of written facts

The route is already limited before overwriting. We test fourteen fact pairs with 47 questions, each answerable only with both facts, screened so that the original model fails every question and a model with both facts in its prompt answers most (Appendix A). Each pair’s facts are placed in the weights or context in all four combinations, using study writes at 192 steps per fact and three seeds. Sequential and joint writing give the same result.

Joint use collapses when both facts are written (Figure 10). Over all fourteen pairs, questions requiring both facts are answered at 32% when both facts are in the weights, against 91% when both are in the prompt (paired difference -58.1 points, 95% CI $[-76.8, -35.4]$). Individually unusable facts do not explain the collapse: 87.5% of placements pass an individual-usability screen, which we report as a check rather than apply as a filter.

The bottleneck is self-retrieval. Asked simply to state a fact it was trained on and can answer questions about, the model produces the correct content 34% of the time (8% verbatim, the rest paraphrase), and its failures include inverted relations and blends of the pair’s two facts. Feeding the model’s own retrievals back to it as context accordingly closes only a sixth of the deficit. Written knowledge is reachable by questions that happen to route to it, but the model has no handle it can pull itself. There is nothing to search.

Supplying the true statements in context recovers accuracy to 63%, and a control explains the remaining gap to the 91% reference: a model that wrote one unrelated fact drops to 66% on the same task. The residual gap is generic damage that any write inflicts on fragile multi-step in-context reasoning, not an interaction between the written and contextual copies of a fact. In the bare-statement condition this generic damage is total. After a single unrelated bare-statement write, the model answers every question by reciting its one written statement and scores zero.

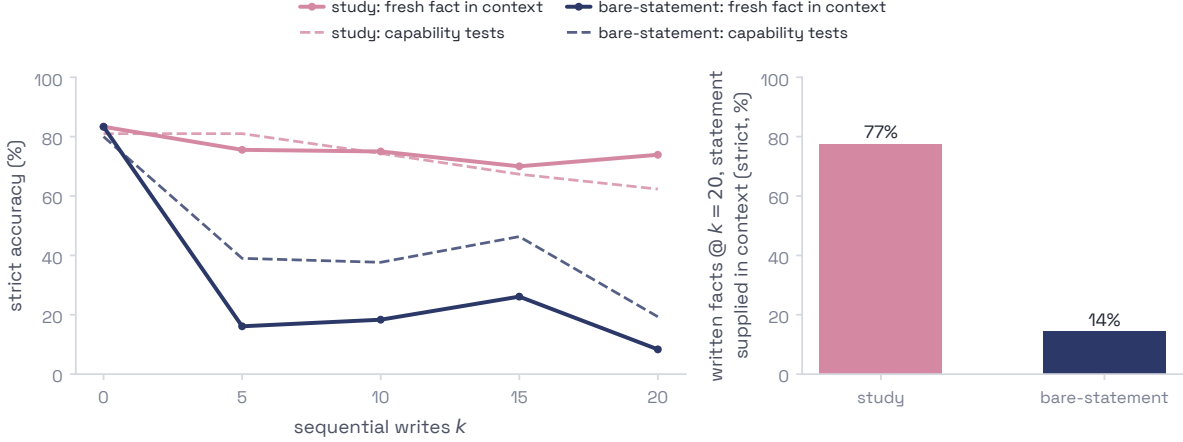


Figure 11: Continued writing does not preferentially erode in-context use. Left: strict accuracy on fresh facts placed in context (solid) and on the held-out capability tests (dashed), across the sequential runs. Right: the sequence’s own written facts at $k=20$, answered with their statements supplied in context.

5.3 In-context use under continued writing

The model’s ability to use context survives the same writes. On the reconstructed checkpoints of the study sequence, accuracy on fresh facts placed in context falls from 83% to 74% over twenty writes, which is less than the held-out capability tests fall on the same models: the difference-in-differences is +9.2 points (95% CI [+3.1, +15.3]), and a positive value means in-context use eroded less than the capability tests (Figure 11). The bare-statement sequence collapses on both measures, so its difference is uninterpretable under the validity screen; the same fresh-fact questions falling to 8% there show that the measurement detects erosion where erosion exists.

The sequence’s own forgotten facts complete the argument. With their statements supplied in context, questions about them return to 77–80% in the study condition, against 14% in the bare-statement condition, whose general collapse extends to this task too. This is not a reading of the stored copy, since a model that had never written the fact would answer a supplied statement equally well. It shows two narrower things: the failure is confined to the route from questions to the stored fact, and the stale written copy does not interfere with a supplied one. As a remedy, supplying the statement concedes the point, because the fact must already be held outside the weights.

6. What repeated writes cost

Continual writing also requires general abilities to survive. We measure capability loss, distributional drift, and retention across twelve conditions: two objectives, adapter ranks 4 and 16, and three writing regimes (sequential, batch at 100 facts, and batch at 200). Sequential conditions merge 20 one-fact adapters; batch conditions train one adapter on all facts at once. The objectives are study-data SFT and context distillation (Snell et al., 2022; Lampinen et al., 2025). Batch distillation uses the original model as teacher, while these sequential conditions use the current merged model; a frozen-teacher sequential variant follows in §6.2. Capability is measured by 100 rule-scored tests, on which the original model scores about 80%. Drift is the KL divergence of next-token distributions from the original model π_0 , averaged over a fixed pool $\mathcal{P}$ of held-out prompts,

$$\text{drift}(\theta) = \mathbb{E}_{x \sim \mathcal{P}} \frac{1}{|x|} \sum_t \text{KL}(\pi_\theta(\cdot | x_{<t}) \| \pi_0(\cdot | x_{<t})), \quad (6.1)$$

the per-token $\text{KL}(\pi_\theta \| \pi_0)$ on text neither model was trained on.

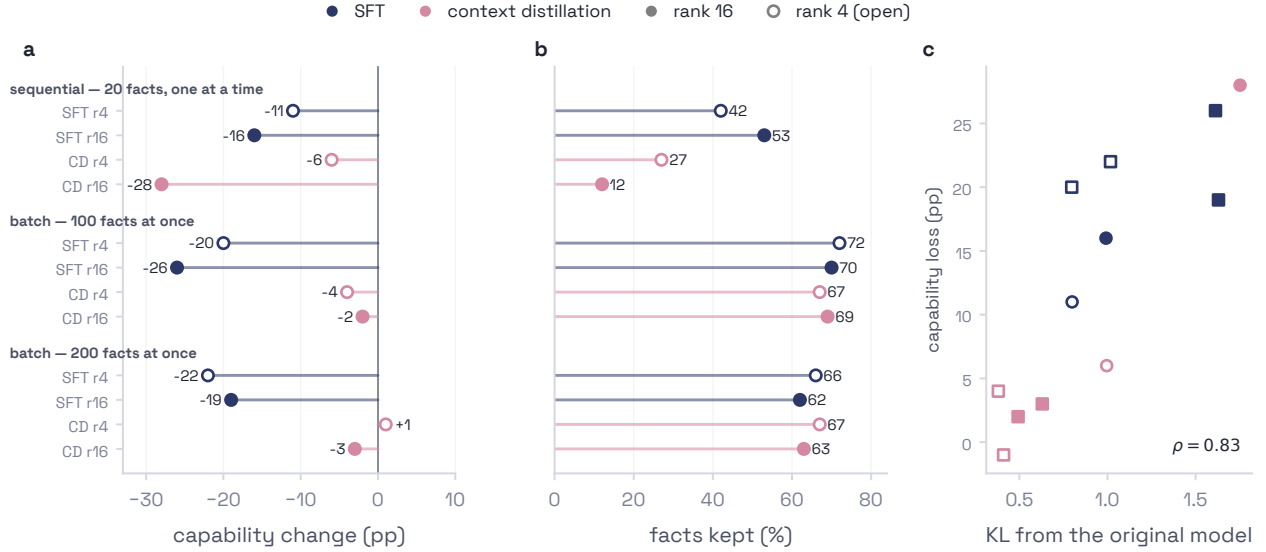


Figure 12: The twelve continual-learning conditions, by regime, objective, and rank. **a**, capability change in percentage points from the $\approx 80\%$ baseline. **b**, facts kept, measured as earlier-fact accuracy after the twentieth sequential write and as all-fact accuracy for the batch conditions. The two extremes are both context distillation. **c**, capability loss against KL divergence from the original model, one point per condition; circles are sequential conditions and squares batch. Accuracy immediately after writing is uniform (76–78% sequential, 62–72% batch) and omitted.

6.1 Capability loss and distributional drift

Across the twelve conditions, capability loss follows KL divergence from the original model (Spearman $\rho = 0.83$; Figure 12c). Objective, rank, and regime matter to the extent that they move the output distribution. This extends the finding of Shenfeld et al. (2025), that KL from the base policy predicts forgetting, to knowledge writing.

The factorial of §3.3 strengthens this ordering on a unique 400-item held-out capability set. Across all sixteen conditions, including the censored one, whose drift and capability measurements remain valid, damage after twenty writes correlates with endpoint KL from the original model at $\rho = 0.946$ (95% CI [0.787, 0.991]) and with cumulative local KL at $\rho = 0.909$ (Figure 4c–d). This is a cost coordinate that holds across methods, not a sufficient causal mechanism: the penalty experiment of §6.3 preserves capability without systematically lowering measured KL from the original.

6.2 The distillation teacher

The extremes of Figure 12 are the same objective. Batch distillation from the original model as teacher writes 200 facts at -3 to $+1$ points of capability, at a third to half the KL of any SFT condition; at rank 16 it beats SFT by 16 points, and the direction holds in all four batch conditions. Sequential distillation, whose teacher is the model’s own accumulated merges, amplifies its own drift (KL 1.75 at rank 16) into the worst of the twelve conditions: -28 points of capability and 11% retention of earlier facts.

The teacher’s distribution, rather than the distillation loss, provides the stability. The original model pins the student to its distribution; a teacher that moves with each write compounds the drift. Since batch accuracy on the written facts is similar across objectives (62–72%), the frozen teacher writes the same knowledge with much less damage. This agrees with single-task results in which frozen teachers are stable and self-distillation diverges (Ye et al., 2026), and with capability preservation under one-hop self-conditioning on demonstrations (Shenfeld et al., 2026). The failure here arises when distillation is iterated through accumulated writes.

We tested this account sequentially, writing one fact at a time and merging each adapter, while always distilling against a frozen copy of the original model. Everything else matched the own-merges condition. The frozen-teacher run writes twenty facts sequentially at $+2$ points of capability, KL 0.48, and 54% retention, indistinguishable

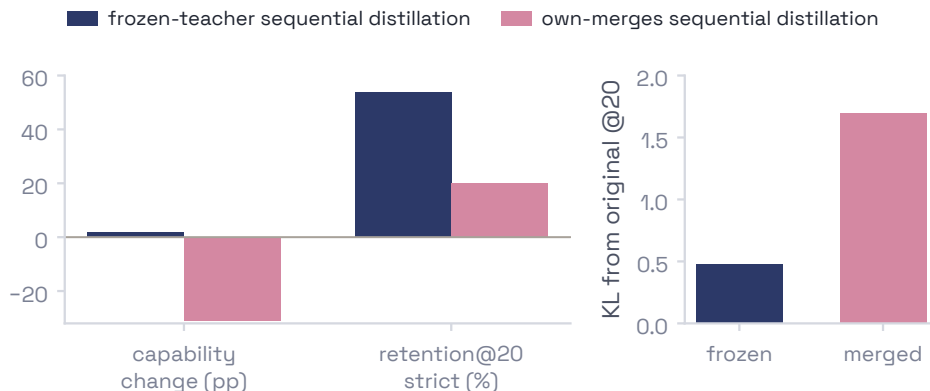


Figure 13: Sequential distillation from a frozen teacher. Distilling each write against a frozen copy of the original model (indigo) preserves capability, retains earlier facts, and stays close to the original. Distilling through the model’s own accumulated merges (rose) loses all three. The own-merges retention bar scores its capped outputs as failures (21%; 34% excluding them).

from the batch condition (Figure 13). The own-merges run loses 31 points, reaches KL 1.70, and retains 21%, or 34% with its capped outputs excluded: the degenerating model loops past any generation budget, so its absolute retention depends on how capped answers are scored. The paired retention difference is 33.0 points (95% CI [24.6, 41.0]). The continual-distillation failure above therefore came from the drifting teacher, not the sequencing. With the original model as a fixed reference, facts can be written online at near-zero cost. No sequential condition we measured does better: the frozen teacher keeps capability intact at retention matching the best fine-tuning condition, which reaches 53% only at -16 points of capability, and even this safest write loses nearly half of its facts by the twentieth.

6.3 An explicit penalty on drift

We added a KL-to-the-current-base penalty to bare-statement SFT, the most damaging write, and swept its weight λ while writing twenty facts sequentially. The penalty is a powerful mitigation (Figure 14, left). Unpenalised, bare-statement sequential SFT collapses the model, losing 66 points of capability with 1% retention; a modest penalty turns this into -19 points and 25% retention at $\lambda = 0.5$ and -5 points and 36% retention at $\lambda = 1.0$, with write accuracy of 53% and 49%. The paired retention gains over $\lambda = 0$ are 23.4 and 35.1 points (95% CIs [16.8, 30.4] and [28.8, 41.6]).

Yet the measured KL from the original remains between 1.8 and 2.4 (Figure 14, right), even as capability recovers by more than sixty points. The penalty preserves abilities without moving the output distribution closer to the original. Thus the $\rho = 0.83$ ordering of §6.1 is robust across methods but is not a sufficient causal handle. The penalty prevents the degenerate collapse of an unpenalised bare-statement sequence; it does not reduce the distance travelled from the original.

6.4 The dissociation of retention and capability

These twelve conditions use their own optimisation budget and question set, so retention levels here are not comparable to those of §4. The twentieth fact is written as reliably as the first: accuracy immediately after writing is 76–78% in all four sequential conditions here, although the heavier 192-step distillation runs of §4.1 instead degenerate by the twentieth write. Earlier facts are another matter. After twenty writes their accuracy is 53% (SFT, rank 16), 42% (SFT, rank 4), 27% (distillation, rank 4), and 12% (distillation, rank 16) (Figure 15).

Two dissociations in Figure 12 resist a single notion of damage. Retention and capability loss disagree: distillation at rank 4 damages capability least among the sequential conditions (-6 points) yet retains earlier facts worse than SFT at the same rank (27% against 42%), while SFT at rank 16 retains best (53%) and erodes capability badly (-16). And higher rank helps retention while hurting capability, the reverse of its usual reputation.

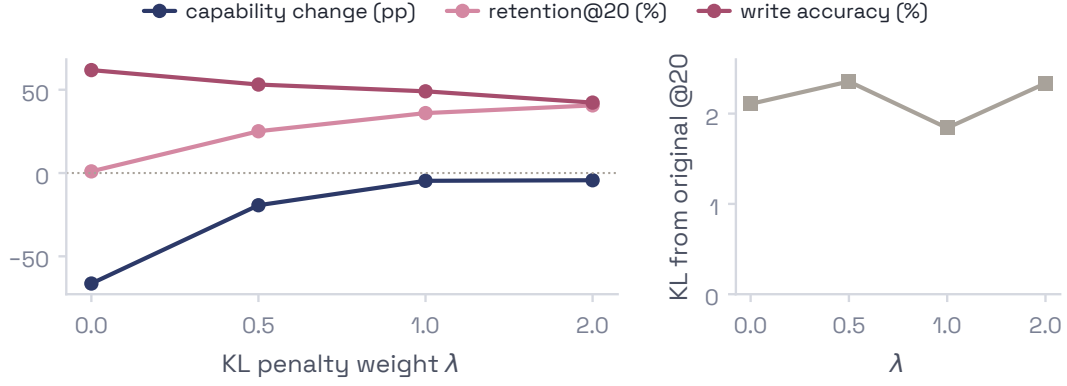


Figure 14: A per-write KL penalty to the current base on bare-statement sequential SFT, swept over its weight λ . Left: capability and retention recover sharply once $\lambda > 0$, while write accuracy falls modestly. Right: the measured KL from the original does not fall systematically with λ , so the rescue is not achieved by reducing drift.

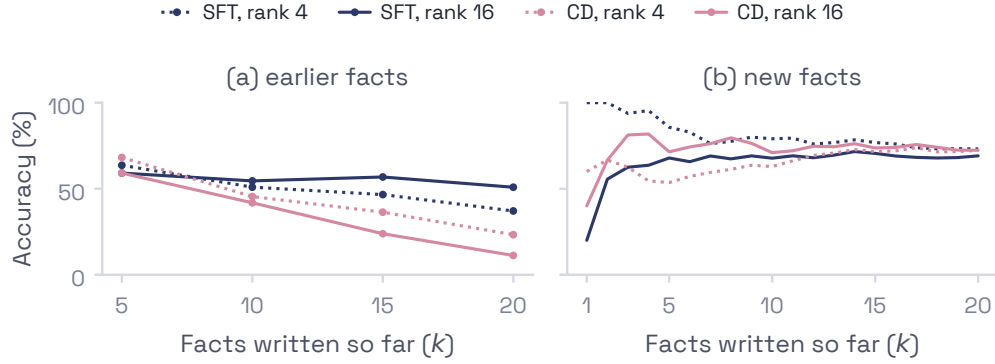


Figure 15: Sequential writing of twenty facts. Left: accuracy on earlier facts after k writes. Right: accuracy on each new fact at the time it is written. The ability to write facts never degrades; the written facts do.

Whatever protects a model’s general abilities is not what protects its previously written facts, which is why survival is a question about the kind of write (§4), not about the amount of damage.

7. Causal tests of creation and interference

The preceding comparisons do not isolate why writing methods differ. We therefore factorise the training data, cross the stored and incoming writes, and test whether local measurements can predict or control interference. Figure 16 summarises the results.

7.1 Training-data factorisation

We trained nine controlled conditions on the same 32 prior-inverting facts, holding the model, adapter rank, learning rate, step count, and seeds fixed. The conditions separately vary recitation against use-bearing examples, narrow against broad prompt coverage, hard labels against teacher distributions, and fixed against resampled trajectories (Appendix C).

Bare-statement training leaves a 27.4-point entailment gap. Training on 24 diverse recall and paraphrase prompts reduces it to 5.4 points, although none contains a derived conclusion. Use-bearing examples in fact scored lower than diverse recitation in this experiment (-8.7 points, 95% CI $[-15.7, -1.9]$). The supported causal variable is therefore prompt breadth, not the presence of an explicit reasoning step. This result does not make every rich objective equivalent; it narrows what is necessary to create usable knowledge.

7.2 Crossed interference

The continual comparisons of §4 change the stored and incoming writes together. A 4×4 cross separates them. Five facts are stored by bare-statement training, study training, online distillation, or offline distillation; fifteen disjoint facts are then written by each method. Retention is measured relative to the baseline after storage, with a second analysis conditioned on successful initial writing.

The incoming-write effect is clear, while the stored-write effect is not. After fifteen later writes, retention changes by -37.0 points for bare-statement writes, against -6.7 for study writes, -1.9 for online distillation, and $+2.4$ for offline distillation. Each contrast with the bare-statement stream excludes zero (30.4, 35.1, and 39.4 points). The stored-write effect changes sign under reasonable definitions of successful writing, so this experiment leaves it open.

A larger 2×2 experiment resolves the ambiguity. It uses 24 new stored facts, three seeds, ten identical later writes, and a common item mask containing only questions answered correctly immediately after both storage methods. The four final retention rates are 9.0% (bare-statement store, bare-statement later writes), 47.8% (bare-statement store, study later writes), 7.8% (study store, bare-statement later writes), and 44.2% (study store, study later writes). Averaged over storage method, changing the incoming writes from bare-statement to study improves retention by 37.6 points (95% CI $[28.9, 46.3]$). The storage-method effect is -2.4 points ($[-9.2, 4.1]$), and the

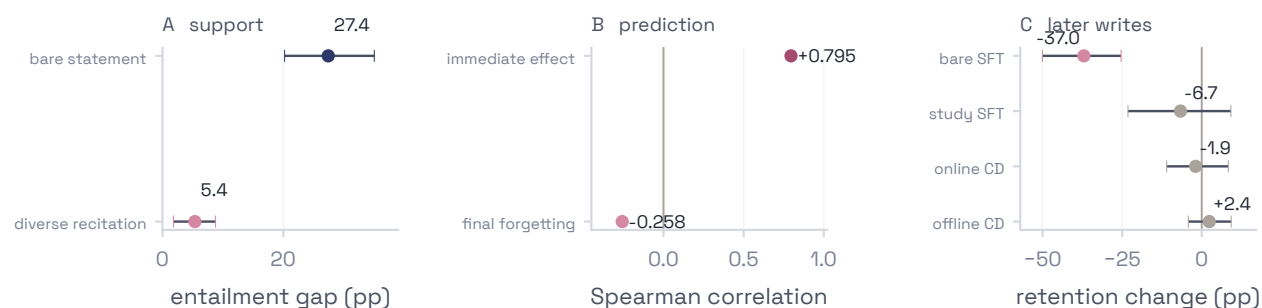


Figure 16: Causal tests of creation and interference. **A:** bare-statement supervision leaves a 27.4-point entailment gap (95% CI $[20.2, 35.1]$); diverse recitation, with no derived conclusions, reduces it to 5.4 points $[1.8, 8.8]$. **B:** the linearised Adam update predicts the next update’s realised effect ($\rho = 0.795$) but not eventual forgetting ($\rho = -0.258$). **C:** change in retention after fifteen later writes, by later-write method, with fact-clustered 95% intervals.

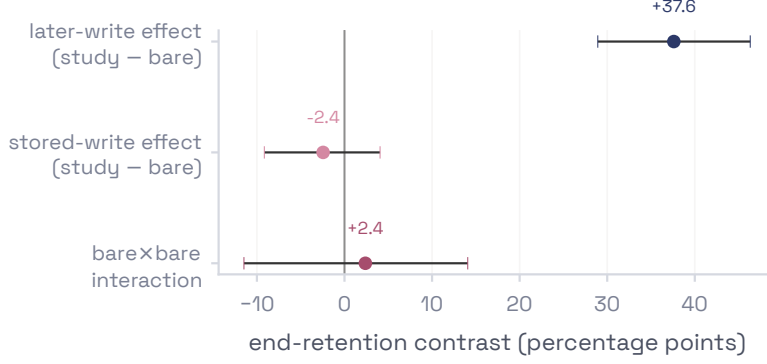


Figure 17: Paired effects on final strict retention in the crossed experiment. Intervals are 95% bootstrap intervals over 24 stored facts after averaging three seeds.

excess loss when bare-statement writes meet a bare-statement store is +2.4 points ($[-11.5, 14.1]$) (Figure 17). Interference is caused by the incoming write, not by how the earlier fact was stored.

This does not show that bare-statement writes selectively erase a stored object. Repetitive generation failures occur in 4.2% and 3.2% of probes under the two bare-statement streams, against 0% under study streams, and under bare-statement streams even the fact-in-prompt reference falls 45.5 points below its level under study streams. Restricting the analysis to the 232 of 250 items without a generation failure in any condition leaves the incoming-write effect at 36.8 points. Generation failures do not create the contrast, but broad generation pathology remains a possible explanation.

7.3 Local prediction and control

We next ask whether interference can be read from an update. At steps 0, 1, 4, 16, 64, and 192, we measured changes on held-out prompts (statement, gold-answer, and default-answer log-probabilities). We also saved the adapter and optimiser, applied one representative update, measured its effect on held-out prompts, and restored the exact state. The candidate early predictor — the step-16 write-use score, the gold-answer gain minus the statement gain on use prompts, frozen before the outcome runs — failed: its rank correlation with final strict-accuracy gain was 0.083.

A linearisation succeeds over the horizon it models. Let g_{use} be the gradient of the earlier facts’ use loss at the current parameters, and $\Delta\theta$ the preconditioned Adam update. Its predicted effect on earlier knowledge is

$$\widehat{\Delta L}_{\text{use}} = g_{\text{use}}^{\top} \Delta\theta. \quad (7.1)$$

The prediction correlates 0.795 with the realised immediate effect but -0.258 with forgetting after the full sequence. It describes the next update, not a fifteen-update trajectory.

None of the three intervention families succeeded. First, adding 48 steps of bridging data (examples designed to connect the recited statement to its conclusions) to 144 bare-statement steps produced no condition that simultaneously halved the entailment gap, gained ten strict-accuracy points, preserved lenient accuracy, and avoided extra capability damage. This is a null for that finite recipe, not for bridging data in general. Second, activation-guided projection, patching, and block swaps did not meet their rescue or kill thresholds: the largest rescue was 5.9 points against a 10-point threshold. These nulls accord with the warning of Hase et al. (2023) that localisation need not identify the best site to edit.

Third, we projected each Adam update away from measured conflict with the earlier facts,

$$\Delta\theta_{\perp} = \Delta\theta - \max(0, \hat{g}_{\text{use}}^{\top} \Delta\theta) \hat{g}_{\text{use}}, \quad \hat{g}_{\text{use}} = g_{\text{use}} / \|g_{\text{use}}\|. \quad (7.2)$$

The protected optimiser improves retention by only 1.4 points over ordinary Adam (95% CI $[-2.1, 5.0]$, five unique stored facts), far below the 15-point target and no better than a norm-matched random projection. Parameter-level gradient mitigation can work elsewhere (Yang et al., 2026); here the measured conflict did not provide causal

control. Appendix C reports every threshold and control. Broad training data changes the functional object, and bare-statement incoming writes destroy access to it, but none of our local handles explains or repairs the long-horizon dynamics.

8. Related work

Fine-tuning learns unknown facts more slowly than known ones and can increase hallucination once new facts are finally learned (Gekhman et al., 2024; Zucchet et al., 2025). Facts held in weights also fail reversals and two-hop compositions that succeed when the same facts appear in the prompt (Berglund et al., 2024; Balesni et al., 2024). Even trained from scratch, two-hop composition over stored facts emerges only under extreme in-distribution overtraining and does not generalise systematically (Wang et al., 2024a). Concurrent work locates this failure mechanistically: memorised facts land in layers that direct recall reaches but mid-layer multi-hop computation does not, and relocating their activations partially restores use (Dai et al., 2026). Data breadth is the established remedy: paraphrases make stored facts easier to extract (Allen-Zhu and Li, 2024), while augmented corpora outperform more documents or tokens (Yang et al., 2025b; Park et al., 2025). We separate optimisation budget from breadth in §3 and isolate diverse recitation in §7.

Knowledge editing encounters the same gap. Locate-and-edit methods can answer single-hop questions above 90% yet fail multi-hop use (Meng et al., 2022, 2023; Zhong et al., 2023; Cohen et al., 2024), and their reported success falls under realistic questioning (Yang et al., 2025a; Xie et al., 2025). Edits can suppress rather than erase an old fact (Holmov et al., 2026), while sequential editing degrades gradually and then catastrophically (Gupta et al., 2024). At 10^5 updates, continual fine-tuning with adapter merging outperforms the tested editors (Thede et al., 2025). Editors that survive many updates instead construct an address or constrain the incoming update (Hartvigsen et al., 2023; Wang et al., 2024b; Fang et al., 2025).

Several findings likewise identify forgetting with lost access. Continual-learning losses can reverse after a few unrelated examples (Zheng et al., 2025); cues can revive abilities no longer elicited by ordinary prompts (Sun et al., 2025b); recall can limit knowledge at pretraining scale (Calderon et al., 2026); and a newly learned fact can prime unrelated generations (Sun et al., 2025a). Human memory research distinguishes available from accessible information (Tulving and Pearlstone, 1966), makes retrieval depend on the match between cue and encoding (Tulving and Thomson, 1973), and attributes forgetting to interference from new learning rather than decay (McGeoch, 1932; Anderson et al., 1994). We use this vocabulary descriptively, without claiming a shared mechanism.

Context distillation internalises documents, prompts, and skills by matching a fact-in-prompt teacher (Snell et al., 2022; Eyuboglu et al., 2025), and can propagate facts to entailed inferences better than direct fine-tuning (Padmanabhan et al., 2023). Frozen teachers are stable where self-distillation can diverge (Ye et al., 2026; Shenfeld et al., 2026); iterating through a moving teacher connects this failure to recursive self-training (Shumailov et al., 2024). Regularisation towards earlier parameters is the classical remedy for forgetting (Kirkpatrick et al., 2017). Recent work instead links forgetting to KL from the base policy (Shenfeld et al., 2025), and finds that low-rank training learns and forgets less (Biderman et al., 2024), with the forgetting attributed to intruder directions rather than rank itself (Shuttleworth et al., 2025). Classical plasticity loss, by contrast, concerns a model losing the ability to learn (Dohare et al., 2024).

The two input channels support different generalisation. Weight-based generalisation is rule-based while context-based generalisation is exemplar-based (Chan et al., 2022), in-context learning is more flexible than fine-tuning on matched content (Lampinen et al., 2025), and retrieval outperforms continued pretraining for injecting new documents (Ovadia et al., 2024). Some systems therefore keep new knowledge in compressed context (Mu et al., 2023) or in key-value memories with explicit addresses (Berges et al., 2024). When context contradicts parametric memory, override probability falls with confidence in the prior (Longpre et al., 2021; Xie et al., 2024; Wu et al., 2024; Xu et al., 2024), and contradictions are harder to instil deeply (Ghosal et al., 2024; Slocum et al., 2025). We manipulate this fact-prior relationship in Appendix D.

Controlled synthetic knowledge supports mechanistic experiments (Allen-Zhu and Li, 2024; Morris et al., 2025; Kirchenbauer et al., 2025; Li et al., 2025; Wang et al., 2026). Judge bias and nondeterminism remain hazards (Zheng et al., 2023; Wataoka et al., 2024; Yuan et al., 2025). Our instrument combines invented entities, five

question types, floor and fact-in-prompt anchors, and generation-time screening; §2.1 describes generative-lineage and checker failures that this design prevents.

9. Discussion

9.1 Stored but question-keyed

The experiments separate creation, storage, access, and destruction. Broad training data creates knowledge that can be used beyond recitation. Later writes can make that knowledge behaviourally unavailable even though checkpoint reconstruction shows that its stored trace remains. The crossed experiments locate the interference in the incoming write rather than in the way the earlier fact was stored. Thus the same-method comparisons show that bare-statement training both creates and preserves knowledge poorly, but do not imply an intrinsic ordering of storage robustness.

A successful weight write is therefore question-keyed. The model can answer the single-fact questions used during training, yet cannot reliably state all of its written facts on demand or bring two of them into a joint computation. The missing capability is what we mean by an *address*: a general handle for reaching a fact beyond the questions that happen to route to it. After further writing, old questions can route to the newest fact instead. Supplying the same content in context provides a general-purpose handle: it supports joint use, survives further writing, and works for forgotten facts as well as fresh ones. Concurrent activation-patching evidence agrees at the mechanistic level: memorised facts sit where recall reaches them but multi-step computation does not, and moving them into the computation’s activations restores use without retraining (Dai et al., 2026). Retrieval-augmented systems preserve this handle outside the weights. The write stores content, while the context supplies its address at use time.

This distinction also explains why the remedies separate. A frozen teacher or an explicit KL penalty can preserve general capabilities, but does not preserve every route to earlier facts. Local measurements can accurately predict the next update without forecasting the eventual trajectory, and projecting away the measured conflict does not improve on a random projection. The remaining causal problem is not merely where a write leaves a trace, but which intervention gives that trace a route that survives subsequent writes.

9.2 Limits and boundaries

All experiments use one base model family, Qwen3-4B, with an 8B replication of the entailment gap. The facts are invented and follow one style. We use LoRA adapters except where full fine-tuning is stated. The causal factorisation covers 32 facts, with repeated measurements rather than independent samples. The crossed experiments cover five and then 24 stored facts, three seeds, and ten to fifteen later writes. Bare-statement streams also degrade fact-in-prompt generation, so their effect is not shown to be selective erasure.

The access experiments cover two training methods at one operating point. The storage probe reads only the statement’s log-probability. The relearning comparison rests on three forgotten facts, and the joint-use evaluation covers fourteen fact pairs. A 60-item audit found seven judge errors; correcting them moves the later-write effect from 37.6 to 37.8 points. The bridge and optimiser nulls apply only to the tested recipes, thresholds, and facts.

The boundary is accumulation, not all writing. Weight updates remain useful as a cache when a canonical copy exists elsewhere, because frozen-teacher batch distillation preserves general capability while consolidating many facts. Pretraining also installs usable knowledge and can be understood as the limit of broad study data. Finally, our evidence concerns discrete facts; whether fine-tuned skills fail in the same way remains untested.

9.3 Continual writing

Continual writing has three requirements. Each write must create usable knowledge, general abilities must survive accumulating updates, and earlier facts must remain reachable. Data breadth makes the first tractable, while a frozen teacher or explicit penalty makes the second tractable. The third remains unresolved. Forgetting reflects lost access rather than erased storage, joint use can fail before later writes arrive, and none of the local signals we

tested controls long-horizon interference. For facts that must be retrieved on demand, composed, and preserved through further training, weights are therefore the wrong system of record: they store content without creating an address for it.

A stronger mechanistic account must connect measurement to control. It should forecast final behaviour beyond the next update, protect earlier facts without blocking new ones, and explain how broad training data connects content to more routes. Objectives may provide another lever. Our reinforcement-learning condition occasionally gained reward without installing the fact, because accuracy alone does not distinguish storage from other ways of producing an answer. An objective that also prices reasoning length might reward direct recall, but whether it would place knowledge in the weights is untested.

The practical guidance is correspondingly narrow. Use broad data for each write, avoid bare-statement training when later updates must preserve existing knowledge, and distil against a frozen copy of the original model. When facts must compose or survive extensive later writing, keep them in context. Models continue to learn new facts after many writes; what fails is the route back to the earlier ones. Continual learning belongs in a channel that creates an address as well as storing content.

References

- Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 3.1, knowledge storage and extraction. *arXiv preprint arXiv:2309.14316*, 2024.
- Michael C. Anderson, Robert A. Bjork, and Elizabeth L. Bjork. Remembering can cause forgetting: Retrieval dynamics in long-term memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 20(5): 1063–1087, 1994.
- Mikita Balesni, Tomasz Korbak, and Owain Evans. Lessons from studying two-hop latent reasoning. *arXiv preprint arXiv:2411.16353*, 2024.
- Vincent-Pierre Berges, Barlas Oğuz, Daniel Haziza, Wen-tau Yih, Luke Zettlemoyer, and Gargi Ghosh. Memory layers at scale. *arXiv preprint arXiv:2412.09764*, 2024.
- Lukas Berglund, Meg Tong, Max Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. The reversal curse: LLMs trained on “A is B” fail to learn “B is A”. *arXiv preprint arXiv:2309.12288*, 2024.
- Dan Biderman, Jacob Portes, et al. LoRA learns less and forgets less. *Transactions on Machine Learning Research*, 2024.
- Nitay Calderon, Eyal Ben-David, Zorik Gekhman, Eran Ofek, and Gal Yona. Empty shelves or lost keys? recall is the bottleneck for parametric factuality. *arXiv preprint arXiv:2602.14080*, 2026.
- Stephanie C. Y. Chan, Ishita Dasgupta, Junkyung Kim, Dharshan Kumaran, Andrew K. Lampinen, and Felix Hill. Transformers generalize differently from information stored in context vs in weights. *arXiv preprint arXiv:2210.05675*, 2022.
- Roi Cohen, Eden Biran, Ori Yoran, Amir Globerson, and Mor Geva. Evaluating the ripple effects of knowledge editing in language models. *Transactions of the Association for Computational Linguistics*, 12, 2024.
- Lu Dai, Ziyang Rao, Yili Wang, Hanqing Wang, Hao Liu, and Hui Xiong. Towards mechanistically understanding why memorized knowledge fails to generalize in large language model finetuning. *arXiv preprint arXiv:2607.08393*, 2026.
- Shibhansh Dohare et al. Loss of plasticity in deep continual learning. *Nature*, 632:768–774, 2024.
- Sabri Eyuboglu, Ryan Ehrlich, et al. Cartridges: Lightweight and general-purpose long context representations via self-study. *arXiv preprint arXiv:2506.06266*, 2025.

- Junfeng Fang, Houcheng Jiang, Kun Wang, Yunshan Ma, Jie Shi, Xiang Wang, Xiangnan He, and Tat-Seng Chua. Alphaedit: Null-space constrained knowledge editing for language models. In *ICLR*, 2025.
- Zorik Gekhman, Gal Yona, Roei Aharoni, et al. Does fine-tuning LLMs on new knowledge encourage hallucinations? In *EMNLP*, 2024.
- Gaurav Ghosal, Tatsunori Hashimoto, and Aditi Raghunathan. Understanding finetuning for factual knowledge extraction. In *ICML*, 2024.
- Akshat Gupta, Anurag Rao, and Gopala Anumanchipalli. Model editing at scale leads to gradual and catastrophic forgetting. In *Findings of ACL*, 2024.
- Thomas Hartvigsen, Swami Sankaranarayanan, Hamid Palangi, Yoon Kim, and Marzyeh Ghassemi. Aging with grace: Lifelong model editing with discrete key-value adapters. In *NeurIPS*, 2023.
- Peter Hase, Mohit Bansal, Been Kim, and Asma Ghandeharioun. Does localization inform editing? surprising differences in causality-based localization vs. knowledge editing in language models. In *NeurIPS*, 2023.
- Ali Holmov, Paul Youssef, Nandi Schoots, and Christin Seifert. One mask to rule them all: On hidden facts after editing and how to find them. *arXiv preprint arXiv:2605.28839*, 2026.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022.
- John Kirchenbauer et al. FictionalQA: A dataset for studying memorization and knowledge acquisition. *arXiv preprint arXiv:2506.05639*, 2025.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 2017.
- Andrew K. Lampinen et al. On the generalization of language models from in-context learning and finetuning: a controlled study. *arXiv preprint arXiv:2505.00661*, 2025.
- Aochong Oliver Li et al. Memorization vs. reasoning: Updating LLMs with new knowledge. *arXiv preprint arXiv:2504.12523*, 2025.
- Shayne Longpre, Kartik Perisetla, Anthony Chen, et al. Entity-based knowledge conflicts in question answering. In *EMNLP*, 2021.
- Kevin Lu and Thinking Machines Lab. On-policy distillation, 2025.
- John A. McGeoch. Forgetting and the law of disuse. *Psychological Review*, 39(4):352–370, 1932.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *NeurIPS*, 2022.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. In *ICLR*, 2023.
- Julian Minder, Clément Dumas, Stewart Slocum, Helena Casademunt, Cameron Holmes, Robert West, and Neel Nanda. Narrow finetuning leaves clearly readable traces in activation differences. *arXiv preprint arXiv:2510.13900*, 2026.
- John X. Morris, Chawin Sitawarin, Chuan Guo, Narine Kokhlikyan, G. Edward Suh, Alexander M. Rush, Kamalika Chaudhuri, and Saeed Mahloujifar. How much do language models memorize? *arXiv preprint arXiv:2505.24832*, 2025.
- Jesse Mu, Xiang Lisa Li, and Noah Goodman. Learning to compress prompts with gist tokens. In *NeurIPS*, 2023.

- Oded Ovadia, Menachem Brief, Moshik Mishaeli, and Oren Elisha. Fine-tuning or retrieval? comparing knowledge injection in LLMs. In *EMNLP*, 2024.
- Shankar Padmanabhan, Yasumasa Onoe, Michael J. Q. Zhang, Greg Durrett, and Eunsol Choi. Propagating knowledge updates to lms through distillation. In *NeurIPS*, 2023.
- Core Francisco Park, Zechen Zhang, and Hidenori Tanaka. New news: System-2 fine-tuning for robust integration of new knowledge. *arXiv preprint arXiv:2505.01812*, 2025.
- Idan Shenfeld, Jyothish Pari, and Pulkit Agrawal. RL’s razor: Why online reinforcement learning forgets less. *arXiv preprint arXiv:2509.04259*, 2025.
- Idan Shenfeld, Mehul Damani, Jonas Hübner, and Pulkit Agrawal. Self-distillation enables continual learning. *arXiv preprint arXiv:2601.19897*, 2026.
- Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759, 2024.
- Reece Shuttleworth, Jacob Andreas, Antonio Torralba, and Pratyusha Sharma. LoRA vs full fine-tuning: An illusion of equivalence. In *ICLR*, 2025.
- Stewart Slocum, Julian Minder, et al. Believe it or not: How deeply do LLMs believe implanted facts? *arXiv preprint arXiv:2510.17941*, 2025.
- Charlie Snell, Dan Klein, and Ruiqi Zhong. Learning by distilling context. *arXiv preprint arXiv:2209.15189*, 2022.
- Chen Sun, Renat Aksitov, Andrey Zhmoginov, Nolan Andrew Miller, Max Vladymyrov, Ulrich Rueckert, Been Kim, and Mark Sandler. How new data permeates llm knowledge and how to dilute it. *arXiv preprint arXiv:2504.09522*, 2025a.
- Huashan Sun, Yizhe Yang, Yinghao Li, Jiawei Li, and Yang Gao. Unveiling and addressing pseudo forgetting in large language models. In *Findings of ACL*, 2025b.
- Lukas Thede et al. Understanding the limits of lifelong knowledge editing in LLMs. In *ICML*, 2025.
- Endel Tulving and Zena Pearlstone. Availability versus accessibility of information in memory for words. *Journal of Verbal Learning and Verbal Behavior*, 5(4):381–391, 1966.
- Endel Tulving and Donald M. Thomson. Encoding specificity and retrieval processes in episodic memory. *Psychological Review*, 80(5):352–373, 1973.
- Wang et al. SynthWorlds: Controlled parallel worlds for disentangling reasoning and knowledge in language models. In *ICLR*, 2026.
- Boshi Wang, Xiang Yue, Yu Su, and Huan Sun. Grokked transformers are implicit reasoners: A mechanistic journey to the edge of generalization. In *NeurIPS*, 2024a.
- Peng Wang, Zexi Li, Ningyu Zhang, Ziwen Xu, Yunzhi Yao, Yong Jiang, Pengjun Xie, Fei Huang, and Huajun Chen. Wise: Rethinking the knowledge memory for lifelong model editing of large language models. In *NeurIPS*, 2024b.
- Koki Wataoka et al. Self-preference bias in LLM-as-a-judge. *arXiv preprint arXiv:2410.21819*, 2024.
- Kevin Wu, Eric Wu, and James Zou. Clashes: Quantifying the tug-of-war between an LLM’s internal prior and external evidence. *arXiv preprint arXiv:2404.10198*, 2024.
- Jiakuan Xie, Pengfei Cao, Yubo Chen, Kang Liu, and Jun Zhao. Revealing the deceptiveness of knowledge editing: A mechanistic analysis of superficial editing. In *ACL*, 2025.

- Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. In *ICLR*, 2024.
- Rongwu Xu et al. Knowledge conflicts for LLMs: A survey. In *EMNLP*, 2024.
- Mutian Yang, Zisen Zhan, Yutong Chen, Haolin Li, Kaiwen Wang, Kaili Zheng, Yuguang Wang, Qi Wang, Jiandong Gao, and Ji Wu. Collaborative parameter learning: Mitigating forgetting via parameter-level gradient analysis. *arXiv preprint arXiv:2601.21577*, 2026.
- Wanli Yang, Fei Sun, Jiajun Tan, Xinyu Ma, Qi Cao, Dawei Yin, Huawei Shen, and Xueqi Cheng. The mirage of model editing: Revisiting evaluation in the wild. In *ACL*, 2025a.
- Zitong Yang, Neil Band, Shuangping Li, Emmanuel Candès, and Tatsunori Hashimoto. Synthetic continued pretraining. In *ICLR*, 2025b.
- Tianzhu Ye, Li Dong, Yutao Wu, Shaohan Huang, and Furu Wei. On-policy context distillation. *arXiv preprint arXiv:2602.12275*, 2026.
- Yuan et al. Understanding and mitigating numerical sources of nondeterminism in LLM inference. *arXiv preprint arXiv:2506.09501*, 2025.
- Junhao Zheng, Xidi Cai, Shengjie Qiu, and Qianli Ma. Spurious forgetting in continual learning of language models. In *ICLR*, 2025.
- Lianmin Zheng et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *NeurIPS*, 2023.
- Zexuan Zhong, Zhengxuan Wu, Christopher D. Manning, Christopher Potts, and Danqi Chen. Mquake: Assessing knowledge editing in language models via multi-hop questions. In *EMNLP*, 2023.
- Nicolas Zucchet, Jörg Bornschein, Stephanie Chan, Andrew K. Lampinen, Razvan Pascanu, and Soham De. How do language models learn facts? Dynamics, curricula and hallucinations. *arXiv preprint arXiv:2503.21676*, 2025.

Appendix A. Apparatus details

A.1 Glossary of conditions

The paper uses one plain name per condition throughout.

defect	direction of bias
training data and evaluation questions shared a generator	inflates conditions trained on generated data
answer-length cap binding for one answer style	deflates distillation conditions
checker ignored negation	inflates, by phrasing style
answer normaliser destroyed numeric signs	false positives on facts built around inverted signs
judge temperature and environments unpinned	irreproducible borderline judgments
judge silently returning no decision on hard items	reshapes comparisons between answer styles even under fail-closed scoring

Table 2: The six instrument defect classes, and the direction each biases.

name	definition
original model (floor)	the base model, with no exposure to the fact
fact in prompt (ceiling)	the same model with the fact statement in its prompt at inference time
bare-statement training	SFT on the fact sentence itself, in two trivial framings, for the stated step budget
study training	SFT on 24 generated items per fact (paraphrases, question–answer pairs, worked implications, contrasts with the violated default)
diverse recitation	SFT on 24 recitation/paraphrase prompts only — no implications or contrasts (creation factorisation)
diverse transformation support	SFT on 24 items that are all implications/contrasts (creation factorisation)
offline context distillation	KL from the student’s next-token distribution to a teacher with the fact in its prompt, on sequences sampled once from the teacher
online context distillation	the same teacher matched on sequences sampled from the student during training, in forward or reverse KL direction
teacher-transcript ablation	plain cross-entropy on the teacher-sampled sequences (the data half of distillation)
sequential conditions	facts written one at a time, each adapter merged before the next fact trains
batch conditions	one adapter trained on all N facts at once
crossed experiment	facts stored by one method, then overwritten by a stream of later writes from another method
the evaluation	the certified five-type question set of §2
held-out capability tests	100 general-ability tests (mathematics, multiple choice, format following), rule-scored with no judge
two-fact question set	the certified joint-use question set of §5.2

A.2 Instrument defect classes

Table 2 lists the six defect classes of §2.1 and the direction each biases. The first five surfaced in a line-by-line audit triggered by an unexplained drop when one condition was re-evaluated on questions from a different model family; the sixth required auditing the judge’s behaviour rather than its decisions. All six are plausible in any fine-tuning evaluation that generates its own training and test data.

A.3 Training

Qwen3-4B, bf16, thinking mode disabled, base weights frozen. LoRA: rank 16 (or 4 where stated), $\alpha = 32$, dropout 0, applied to all seven projection matrices (attention q/k/v/o, MLP gate/up/down). AdamW, learning rate 2×10^{-4} , loss on answer tokens only, optimiser steps counted at batch size one. Single-fact conditions train a fresh adapter per fact. Sequential conditions merge each fact’s adapter into the running model before the next fact trains. Batch conditions train one adapter on all N facts for $24 \times N$ total steps. In context distillation, teacher and

student share weights, the teacher being a forward pass with the adapter disabled and the fact in context. The teacher-transcript ablation trains with plain cross-entropy on the same teacher-sampled sequences, isolating the data half of distillation; its contrasts are in Appendix B.

The larger crossed experiment uses Qwen3-4B with rank-16 LoRA and the same 2×10^{-4} learning rate. Each stored fact and each of ten shared later facts receives 192 optimiser steps. The 288 sequences cross two storage methods, two later-write methods, 24 stored facts, and three seeds. The two data conditions are matched on steps, not tokens or examples. Runs used a frozen dependency lockfile (torch 2.12.1, transformers 5.13.0, peft 0.19.1); the exact base-model revision was not recorded in the run rows.

The study set is 24 items per fact — paraphrases, question–answer pairs, worked implications, and contrasts with the violated default — generated once and reused across all conditions; the generator never sees the evaluation questions.

A.4 Certification gate

An evaluation is used only after passing a scripted gate:

criterion	threshold	failure it catches
floor	< 20%	questions answerable without the fact
ceiling	> 80%	questions unanswerable even with the fact
truncation	< 5% per condition	answer cap binding unevenly
leak audit	$\leq 3/30$	question wording reveals the answer
pass audit	$\geq 24/30$	judge credits wrong or hedged answers

The primary evaluation passed at floor 3%, ceiling 84%, leak 1/30, pass-audit 29/30. Truncation for the original model (23%) and bare-statement training at 24 steps (11%) is reported but not gating; it deflates the floor side of comparisons. Audits run at temperature zero with majority-of-three judgments per item.

A.5 Tier evaluation: certification amendments

Two amendments were adopted, both before recomputation. First, the leak audit returned different counts on identical inputs (5/30, then 3/30) because its sampling temperature was unpinned; audits were pinned to temperature zero with majority-of-three judgments, and confirm-tier questions are audited only for wording that reveals the entity binding, since answerable-from-common-sense is that tier’s definition and the live floor screen controls it. Second, the fact-in-prompt condition initially scored 77–80% on two tiers against an 80% bar: screening questions for original-model failure selects for hardness, keeping a tail of questions that fail even with the fact available. Such questions were removed, the surviving set was frozen before any training run, and the certifying criterion became the surviving fraction, at least 70% per tier (observed: confirm 76.3%, neutral 95.9%, invert 79.5%; floors on surviving questions 0.9–6.2%; 48/1/23 facts per tier left with fewer than four questions).

A.6 Statistics

Bootstrap 95% confidence intervals over facts. Between-condition comparisons are paired per-fact differences; cross-tier comparisons are paired within triples. Multi-seed cells report per-seed values and their mean. Creation-factorisation and crossed-interference intervals cluster repeated measurements by unique fact; 864 creation units correspond to 32 facts. The optimiser analysis uses a different resampling unit: 60 paired observations over store, seed, and stored fact, drawn from five unique stored facts. The larger crossed analysis intersects the questions answered correctly after both storage methods for each fact–seed pair, averages seeds within each of 24 stored facts, and resamples facts with 20,000 bootstrap draws. The thresholds are 15 points for the later-write contrast and 10 points for the stored-write and interaction contrasts. A 60-item stratified endpoint audit found 53/60 judge agreement; applying all seven manual corrections did not change the result.

Appendix B. Supplementary results

B.1 Single-fact conditions with confidence intervals

Accuracy (strict policy) by question type over the 247-fact evaluation, bootstrap 95% CIs over facts. CD is context distillation; the transcript ablation trains with cross-entropy on the same teacher-sampled sequences.

question type	original model (floor)	bare statement / 24 steps	study set 96 steps	transcript ablation	CD	fact in prompt
recall	1 [0,3]	97 [95,99]	88 [84,92]	91 [88,95]	93 [90,96]	99 [97,100]
application	7 [4,9]	52 [47,57]	73 [68,77]	67 [62,71]	75 [71,79]	91 [87,94]
composition	4 [2,6]	40 [34,45]	60 [55,65]	61 [56,66]	70 [65,75]	83 [80,87]
paraphrase	2 [0,4]	95 [92,98]	83 [78,87]	82 [76,86]	83 [78,88]	93 [90,96]
counterfactual	3 [1,6]	23 [17,29]	50 [43,57]	50 [43,57]	56 [49,63]	75 [69,81]

The bare-statement column is 24 steps and the study column 96 steps (each recipe’s operating point). Paired composition differences ($n = 236$): CD – study set = +10.8 [+4.9, +16.5]; CD – fact in prompt = –13.1 [–18.2, –7.8]; transcript ablation – study set = +1.5 [–4.9, +7.4]; CD – transcript ablation = +9.3 [+4.9, +13.8]. Under the strict policy the distillation gain is real and, unlike our earlier lenient-graded estimate, the CD-over-transcript increment excludes zero: the distillation loss, not only the teacher’s sampled sequences, carries the gain.

B.2 Prior strength as a per-fact predictor (exploratory analysis)

The regression that motivated §D. Each fact’s prior strength was scored 0–3 from the original model’s own answers to that fact’s questions (its wrong answers state the belief it holds), for 241 of 244 facts; 78% scored 2 or 3, reflecting the set’s built-in-conflict construction. Outcomes are per-fact accuracies from existing certified records; the statistic is tie-adjusted Spearman ρ with bootstrap CIs.

outcome	ρ [95% CI]	n
counterfactual accuracy, bare-statement 24	+0.01 [–0.13, +0.16]	179
study 96 / transcript / CD / fact in prompt	+0.02 / –0.06 / –0.12 / –0.04	179
recall accuracy (control), bare 24 / study 96	–0.01 / –0.02	234
retention after 20 writes, four sequential conditions	–0.24 / –0.05 / –0.26 / –0.47	19 each
retention, per-fact mean over conditions	–0.27 [–0.66, +0.21]	19

Prior strength does not predict which facts win at inference time, in any condition. The retention correlation is negative in all four sequential conditions and unpowered at $n = 19$. The powered test on the tier facts subsequently ran (§4.2) and reversed the prediction: prior-inverting facts survived better than neutral ones, which we read post hoc as distinctiveness protecting. The relaxation reading of the prior-as-opponent account, in which sequential training lets the prior recover its most-contradicted facts first, did not survive that test. Two artifacts could have produced the inference-time null here: the compressed conflict range, and a per-fact outcome that is nearly one binary question. The tier design removes both.

B.3 A post-scope-change extension to one hundred writes

The hundred-write extension of Figure 8 covers a balanced eight-cell comparison, reduced from a planned twelve after partial records existed: SFT and offline distillation with bare-statement or self-study data and either low-rank or full-parameter updates. Each tuned operating point has three seeds and one hundred sequential writes. It is a descriptive extension, not a confirmatory factorial: the reduced scope, endpoint contrasts, and uncertainty rule were not fixed before the data existed, and online-distillation partials are excluded.

All 24 retained records pass coverage validation. This experiment’s first-sentence response protocol, which scores only the first sentence of each response, reached a 77.3% in-context ceiling, below the > 80% certification

gate; the miss was accepted before the runs and caps what the endpoint levels can show. At $k = 100$, bare-statement SFT retains 8.6–11.7% of earlier facts and study SFT 33.0–37.7%. Offline distillation retains 25.0–29.4% with bare-statement data and 38.5–43.7% with study data. These are ranges across the low-rank and full-parameter cells, not confidence intervals. Offline distillation is higher in all four matched endpoint comparisons, but the trajectories cross at earlier checkpoints.

Held-out capability losses at the endpoint range from 73.5 percentage points for full-parameter bare-statement SFT to 9.0 points for low-rank study distillation. Full-parameter bare-statement distillation loses 56.9 points; the corresponding low-rank bare-statement SFT and distillation conditions lose 34.2 and 28.7 points. Learning rates were selected by condition, and full versus low-rank updates change both learning rate and parameterisation, so these numbers describe the tested operating points rather than independent causal effects of data, objective, or update parameterisation.

Appendix C. The causal experiments: design, controls, and limits

C.1 Creation factorisation and checkpoints

All creation cells use Qwen3-4B with rank-16 LoRA, learning rate 2×10^{-4} , 192 optimiser steps, three seeds, and the same 32 held-out prior-inverting facts. The nine cells are: two narrow bare-statement prompts; 24 diverse recall and paraphrase prompts with no derived conclusion; two repeated one-step transformation prompts; 24 diverse application, counterfactual, and composition prompts; the study data; hard-label cross-entropy on frozen in-context-teacher rollouts; forward KL on those fixed teacher trajectories; forward KL on trajectories sampled once from the student initialised from the original model; and forward KL on adaptively resampled current-student trajectories. Checkpoints at steps 0, 1, 4, 16, 64, and 192 record gold, statement, teacher, and default-answer log probabilities as well as generated strict and lenient accuracy. The 864 fact-condition-seed units are repeated measurements on 32 facts, not 864 independent facts; intervals cluster by fact.

The micro-update test snapshots both adapter and optimiser state, performs one actual update, measures before-after probe log probabilities, and restores the state byte-for-byte. Its linear predictor includes the realised update magnitude and Adam preconditioning. Generation used a separately certified exact-decode cycle, and judge abstentions are never imputed.

C.2 Crossed interference design and protected optimiser

The 4×4 crossed design stores five facts using bare-statement training, study training, online forward-KL context distillation, or offline forward-KL context distillation, then writes fifteen disjoint facts over them using each of the same methods. Every distillation condition uses a frozen original-model teacher. The stored facts are re-evaluated after 1, 5, 10, and 15 later writes. The primary endpoint is change from the baseline taken after the stored facts were written; a second analysis conditions on successful initial writing. The distinction matters: the stored-write main effect changes under reasonable definitions of successful writing, while the later-write effect does not.

The protected optimiser retains one held-out use prompt and gold answer for each earlier fact. Periodically it aggregates the old-use gradient, forms the actual preconditioned Adam update, and removes the component that increases old-use loss to first order. Ordinary Adam, norm-matched random projection, and replay-free magnitude-based module masking are controls.

C.3 Intervention scope and exploratory observations

Figure 18 shows why the bridge and optimiser gates failed. The selected $k = 8$ bridge condition was retained only as a diagnostic because selection and evaluation were not independent; it halved the gap and gained 11.3 strict-use points, but lost 7.0 lenient points and therefore could not pass the conjunction. No conclusion extends beyond the tested 144+48-step mixture, the candidate examples, and these facts. For the optimiser, targeted projection is also indistinguishable from random projection (difference -0.4 points $[-4.4, 3.6]$). Magnitude masking has the largest estimate, $+5.0$ points $[1.4, 8.9]$, but still misses the target.

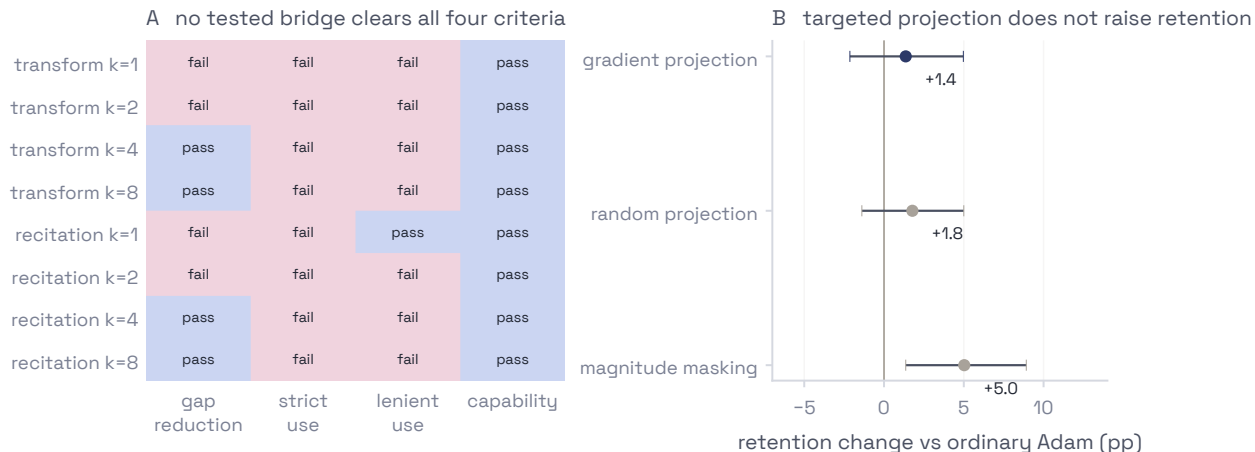


Figure 18: Intervention controls. **A:** Each row is an eligible tested 144-step bare-statement plus 48-step bridge condition; indigo marks a passed individual criterion. No row passes all four. **B:** Retention change relative to ordinary Adam. Targeted gradient projection gains 1.4 points, random projection 1.8, and magnitude masking 5.0. Intervals resample the 60 paired store–seed–fact units; only five stored facts are unique.

Exploratory work found some SFT-written facts decodable at intermediate depth, and restricting updates to deeper layers affected SFT and distillation differently. Our projection, patching, and fixed-block swaps did not establish necessity or sufficiency (activation differences can trace narrow fine-tuning (Minder et al., 2026)), so we retain the depth and logit-lens observations as motivation only: they do not establish a lookup-versus-computation mechanism, and we did not apply a richer lens to components that failed the causal gate.

C.4 Scope and adversarial review

All confirmatory results are scoped to Qwen3-4B, rank-16 LoRA, the 32 creation facts, five stored facts, and fifteen later writes. Each report passed automated balance and provenance checks and independent adversarial review; the main caveats are repeated measurements on a small fact set, sensitivity of the store effect, post-selection in diagnostic bridge rows, and the gap between a one-step linear approximation and a multi-update trajectory. These experiments reject the early-predictor, transformation-necessity, and gradient-projection hypotheses under their protocols. They do not show that all predictors, bridge recipes, internal mechanisms, or optimiser protections must fail.

Appendix D. Prior conflict as the experimental variable

This appendix reports the prior-conflict experiment referenced in §4 and §9. Its primary contrast was inconclusive, and its cue analysis corroborates §3: cuing a question reduces it to the recitation a bare-statement write installs.

Every fact in the primary evaluation contradicts a prior belief by construction, so the results of §3 cannot say whether the prior causes the diversity requirement, while outside evidence says the prior matters (Wu et al., 2024; Slocum et al., 2025). We therefore built a second certified evaluation in which the fact–prior relationship is the manipulated variable (Figure 19).

A regression on the existing facts was not enough: per-fact prior strength gave a null for counterfactual use ($\rho \approx 0$ in every condition, $n = 179$), but that analysis is doubly confounded, by the set’s compressed conflict range and its nearly binary per-fact outcome (Appendix B.2). Manipulating conflict directly, down to zero, removes both problems.

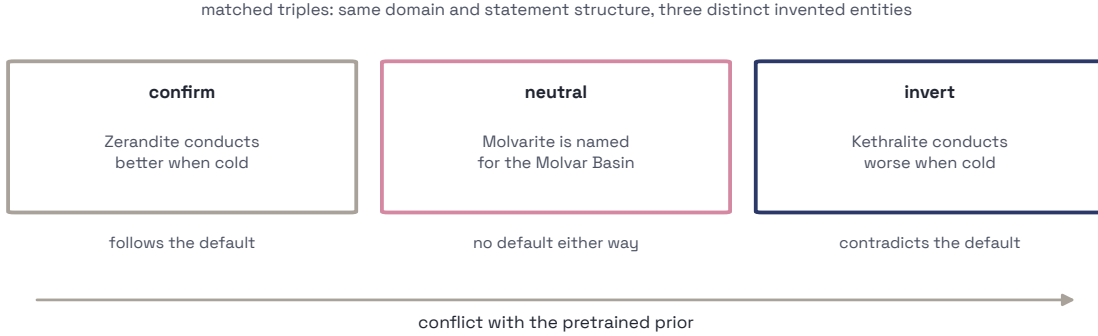


Figure 19: The matched-triple construction. Three invented entities share a domain and statement structure and differ only in how the stated property relates to a prior belief: it confirms the default, is neutral to any default, or inverts the default.

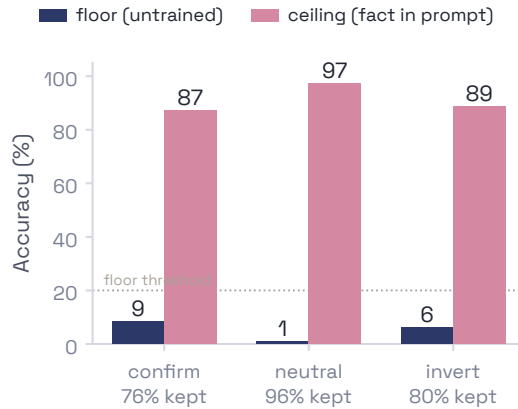


Figure 20: Floor and ceiling for the three tiers under the certified judging pipeline, with the fraction of questions surviving the screen in the tick labels. Floors of 0.9–6.2% show the questions do not leak their answers in any tier.

D.1 Construction

Facts come in matched triples sharing a domain, statement structure, and entity-name style, with three distinct invented entities. The members differ only in how the stated property relates to a prior belief: it *confirms* the default (“Zerandite conducts current more efficiently as its temperature drops toward absolute zero”), is *neutral* to any default (“Molvarite is named after the Molvar Basin where it was first mined”), or *inverts* the default (“Kethralite conducts current less efficiently...”). Entity novelty is common to all three, so the original model fails recall everywhere; only the conflict varies. Matching makes cross-tier comparisons paired. A generation-time screen discarded triples whose tier a fresh model call could not recover blind (47% discarded); after deduplication the set is 80 triples, 240 facts.

Questions follow §2, written by a second model family and kept only if the original model fails them, with two declared asymmetries. Confirm-tier questions about *using* the property are answerable from the default alone, so that tier plays a supporting role; the primary contrast is neutral versus invert. And the fifth question type must differ by tier, since “side with the fact against the default” requires a default: invert keeps counterfactual questions, neutral gets distractor questions (commit to the arbitrary value against a plausible alternative), confirm has four types.

Certification required two amendments, detailed in Appendix A; all three tiers then passed the gate, and the screened question set (76–96% of questions surviving per tier) was frozen before any training run.

A further observation: the fact-in-prompt ceiling itself orders by tier (neutral 96%, invert 80%, confirm 77%). Content that interacts with a prior belief, in either direction, taxes even in-context use.

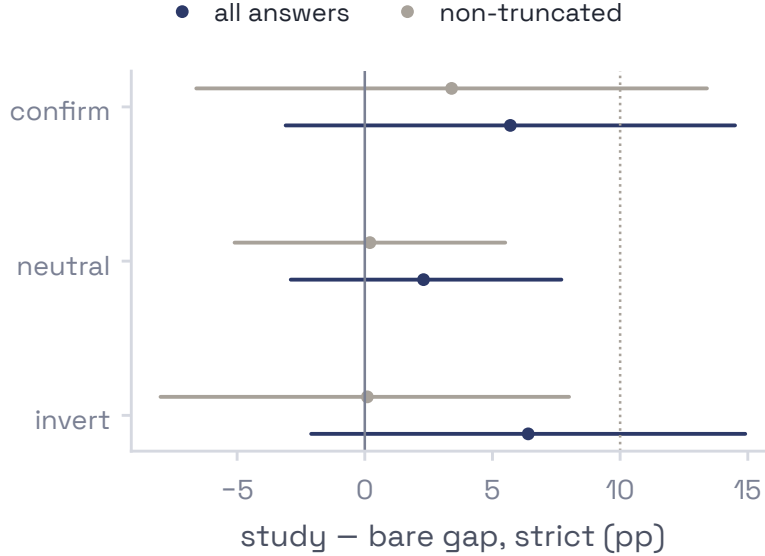


Figure 21: The conflict experiment’s primary quantity (strict policy): the study advantage over bare-statement training on use-type questions, paired within triples, averaged over three seeds, with 95% CIs, by prior tier (invert 77 triples, neutral 80, confirm 61). Every interval crosses zero and none reaches the +10 pp support band (dotted); recomputed on non-truncated rows only (“non-truncated”), the gaps collapse toward zero — the residual signal is a truncation artefact.

D.2 The conflict experiment

If the prior causes the diversity requirement, the study-training advantage on use-type questions should collapse for neutral facts and persist for inverting facts. If diversity instead works by making the fact retrievable under many phrasings, the advantage should persist in both.

The design: bare-statement and study training at 24 and 96 steps, on all 240 facts, three seeds, scored on the frozen question set by the certified pipeline. The primary quantity is the study–bare gap at 96 steps on use-type questions (application, composition, and the tier’s fifth type), paired within triples, averaged over seeds, per tier. Recall questions are the negative control: neither account predicts a tier-by-recipe interaction on recitation, so one appearing would indicate an instrument artefact. We also examine the composition gap to the ceiling by tier — if neutral facts compose near their 96% ceiling while inverting facts stay capped, the composition limit of §3 is itself a conflict phenomenon.

D.3 Results

The full grid comprises 13,728 records (240 facts $\times$ 4 recipes $\times$ 3 seeds), all scored on the frozen question set by the certified pipeline.

Under the strict policy the gaps are small, positive, and every confidence interval crosses zero (Figure 21); the paired invert–neutral interaction is +4.3 pp [−6.1, +14.2]. Against the support criteria (an invert gap $\geq$ +10 pp with the interval excluding zero, and a neutral gap $\leq$ +5 pp) the result is **inconclusive**: this instrument neither reproduces the primary evaluation’s diversity effect nor rules it out. One material caveat explains part of the residual: bare-statement conditions truncate far more at the 512-token answer cap than study conditions (6.1% versus 0.1% at 96 steps), the same defect class as the answer-length cap of §2.1. Recomputed on non-truncated rows only, every gap collapses toward zero (invert +0.1 pp, neutral +0.2 pp), so the small positive signal in the unfiltered values is largely a truncation artefact working for the diversity hypothesis. The planned remedy is more triples; the sharper one, developed below, is fixing the question style that drives the discrepancy. The negative control behaved: recall gaps are uniform across tiers (−6.2/−5.4/−8.7 pp, the familiar study-lags-recitation pattern of §3), with no tier-by-recipe interaction.

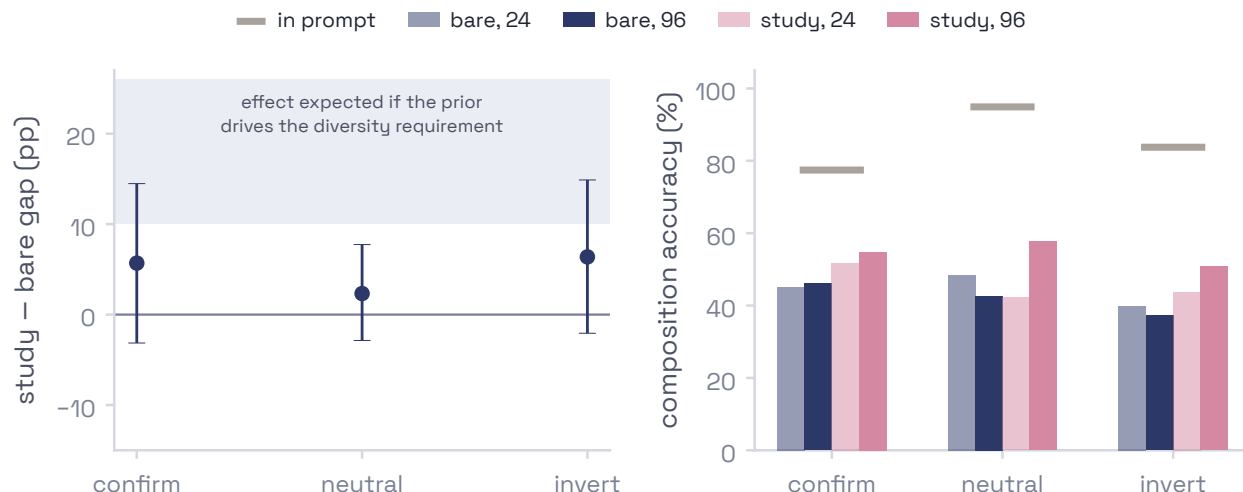


Figure 22: Left: under the strict policy the diversity effect is small and every interval crosses zero; the expected effect size (shaded band) sits at the upper edge of the intervals. Right: composition sits at 37–58% for every recipe in every tier while the in-context ceiling spans 77–95%.

D.4 The cue mechanism, measured

The inconclusive result has a specific cause, and the dual-policy regime makes it visible. On the tier counterfactual questions bare-statement training scores 87–88% strict (ceiling 93%); on the primary evaluation’s counterfactual questions the same recipe scores 23% strict. Study training is comparatively stable across the two (79% on the tiers, 50% on the primary). The gap is not prior strength: under the identical scorer the inverting tier facts carry *more* prior conflict than the primary set (89% versus 78% high-conflict). What separates the two instruments is question style. The tier counterfactual questions *cue the contrast* in their wording (“Where many pigments visibly bleach...” — the surface form of the trained statement), while the primary evaluation presents bare scenarios that demand *spontaneous override*.

Read through the entailment gap of §3.1, this is one mechanism, not two. A cued question does for free the step that diverse data otherwise has to buy: it converts reciting the trained statement into stating the asked-for conclusion. Bare-statement training, which installs recite-plus-one-step knowledge, therefore looks strong exactly where the question supplies the missing step (the cued tiers, high strict scores) and weak where it does not (the uncued primary, low strict scores); study training, which installs stated conclusions, is roughly cue-independent. Because the two instruments differ in more than their cuing, we ran a controlled confirmation: a full cue × recipe design, retrained on both fact sets with cued and uncued counterfactual question sets, each verified by a blind cue-classifier. It splits the effect in two. The *cue main effect* is robust, large, and specific to bare-statement training: stating the default and asking for the contrast raises bare-statement strict accuracy 52 → 88% on the tier facts and 23 → 99% on the primary facts, roughly twice the lift it gives the study condition (difference-in-differences +24.7 pp and +44.1 pp). The mechanism is not that cuing unlocks applicable knowledge. The cued questions are, by construction, answerable by *reciting* the written statement: “normally *X*; for this entity, what instead?” hands the model the frame. Cuing therefore converts bare-statement training’s recitations from insufficient to sufficient, collapsing its entailment gap from 31.8 pp uncued to 0.9 pp cued. It rescues bare-statement knowledge by reducing the question to the recitation the write installed (§3), not by making that knowledge more usable. The magnitude is further inflated by the cued questions’ shorter inference distance, with the fact-in-prompt condition scoring 95% cued against 76% uncued. But the *diversity* × *uncued* interaction — the claim that study training uniquely buys spontaneous override — did not reproduce on the tier facts under bare uncued phrasing. A diagnostic resolved why. The tier uncued question set, though it hides the default, was about half *recitation-answerable*, meaning the written sentence, restated, is the answer; the primary questions are ~2%. Recitation specifically flatters bare-statement training: its item-level pass is predicted by recitability (point-biserial $r = 0.42$), and study training’s is not ($r = -0.22$). Restricted to application-only tier items — where reciting the sentence does not

answer the question — the diversity advantage reappears, matching the primary facts. So the diversity requirement for *spontaneous* (uncued, application) override stands on the primary facts and is direction-consistent on the tier facts; a clean tier replication awaits a purpose-built application-only question set. We treat the cue main effect as the confirmed result here and scope the diversity claim to spontaneous-application questions.

The composition limit is conflict-independent. Composition sits at 37–58% for every tier and recipe while the ceiling ranges from 77% to 95%, and the gap is *largest* at zero conflict (neutral: 58% versus a 95% ceiling). Conflict joins training data, training objective, update method, and model scale as manipulations this limit survives. Composition failure is about where the fact lives — weights versus activations — not about what it fights.